SCIENCE

The celestial worlds discover'd : or, conjectures concerning the inhabitants, plants and productions of the worlds in the planets

THE Celeſtial Worlds DISCOVER’D: OR, CONJECTURES Concerning the INHABITANTS, Plants and Productions OF THE Worlds in the Planets.

Written in Latin by CHRISTIANUS HUYGENS, And inſcrib’d to his Brother CONSTANTINE HUYGENS Late Secretary to his Majeſty King William.

The Second Edition, Corrected and Enlarged.

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TO THE READER.

This Book was juſt finiſhed, and deſigned for the Preſs, when the Author, to the great loſs of the Learned World, was ſeized by a Diſeaſe that brought him to his Death. However he took care in his laſt Will of its Publication, deſiring his Brother, to whom it was writ, to take that Trouble upon him. But he was ſo taken up with Buſineſs and Removals, (as being Secretary in Holland to the King of Great Britain) that he could find no time for it till a Year after the Death of the Author: When it ſo fell out, that the Printers being ſomewhat tardy, and this Gentleman dying, the Book was left without either Father or Guardian. Yet it now ventures into the Publick, in the ſame Method that it was writ by the Author, and with the ſame Inſcription to his Brother, tho’ dead; in confidence that this laſt Piece of his will meet with as kind a Reception from the World as all the other Works of that Author have. ’Tis true there are not every where Mathematical Demonſtrations; but where they are wanting, you have probable and ingenious Conjectures, which is the moſt that can be reaſonably expected in ſuch matters. What belongs to, or has any thing to do with Aſtronomy, you will ſee demonſtrated, and the reſt ingeniouſly and ſhrewdly gueſs’d at, from the Affinity and Relation of the heavenly Bodies to the Earth. For your farther Satisfaction read on, and farewel.

THE PUBLISHER TO THE READER.

I Doubt not but I ſhall incur the Cenſures of learned Men for putting this Book into Engliſh, becauſe, they’ll ſay, it renders Philoſophy cheap and vulgar, and, which is worſe, furniſhes a ſort of injudicious People with a ſmattering of Notions, which being not able to make a proper uſe of, they pervert to the Injury of Religion and Science. I confeſs the Allegation is too true: but after Biſhop Wilkins, Dr. Burnet, Mr. Whiſton and others, to ſay nothing of the ancient Philoſophers, who wrote in their own Tongues; I ſay, after theſe great Authors have treated on as learned and abſtruſe Subjects in the ſame Language, I hope their Example will be allowed a ſufficient excuſe for printing this Book in Engliſh.

Concerning this Edition I can ſay, that I have taken care to have the Cutts exactly done, and have placed each Figure at the Page of the Book that refers to it, which I take to be more convenient to the Reader than putting them all at the End.

I have been careful to procure the beſt Paper, that I might in ſome meaſure come up to the Beauty of the Latin Edition, though this bear but half the Price of it.

And I hope the Tranſlator has expreſſed the Author's Senſe aright, and has not committed Faults beyond what an ingenuous Reader can pardon.

NEW CONJECTURES Concerning the Planetary Worlds, THEIR INHABITANTS AND PRODUCTIONS.

Written by Chriſtianus Huygens, and inſcribed to his Brother Conſtantine Huygens.

BOOK the Firſt.

A Man that is of Copernicus's Opinion, that this Earth of ours is a Planet, carry'd round and enlighten'd by the Sun, like the reſt of the Planets, cannot but ſometimes think, that it's not improbable that the reſt of the Planets have their Dreſs and Furniture, and perhaps their Inhabitants too as well as this Earth of ours: Eſpecially if he conſiders the later Diſcoveries made in the Heavens ſince Copernicus's time, viz. the Attendants of Jupiter and Saturn, and the champaign and hilly Countries in the Moon, which are a ſtrong Argument of a Relation and Kin between our Earth and them, as well as a Proof of the Truth of that Syſtem. This has often been our Talk, I remember, good Brother, over a large Teleſcope, when we have been viewing thoſe Bodies, a Study that your continual Buſineſs and Abſence have interrupted for many Years. But we were always apt to conclude, that 'twas in vain to enquire after what Nature is doing there, ſeeing there was no likelihood of ever coming to any Certainty of the Enquiry. Nor could I ever find that any Philoſophers, either antient or modern, have attempted any thing upon this Subject. At the very Birth of Aſtronomy, when the Earth was firſt aſſerted to be Spherical, and to [Some have already talk'd of the Inhabitants of the Planets, but went no farther.] be ſurrounded with Air, even then there were ſome Men ſo bold as to affirm, there were an innumerable Company of Worlds in the Stars. But later Authors, ſuch as Cardinal Cuſanus, Brunus, Kepler, (and if we may believe him, Tycho was of that opinion too) have furniſhed the Planets with Inhabitants. Nay, Cuſanus and Brunus have allowed the Sun and fixed Stars theirs too. But this was the utmoſt of their Boldneſs; nor has the ingenious French Author of the Dialogues about the Plurality of Worlds carried this Matter any farther. Only ſome of them have coined ſome Stories of the Men in the Moon, juſt as probable as Lucian's true Hiſtory; among which I muſt count Kepler's, which he has diverted us with in his Aſtronomical Dream. But a while ago thinking ſomewhat ſeriouſly of this matter (not that I count my ſelf quicker-ſighted than thoſe great Men, but that I had the Happineſs to live after moſt of them) the Enquiry appeared not ſo impracticable, nor the Way ſo ſtopt up with Difficulties, but that there was very good room left for probable Conjectures. As they came into my Head, I put them down into common Places, and ſhall now try to digeſt them into ſome Method for your better Conception of them, and add ſomewhat of the Sun and fix'd Stars, and the Extent of that Univerſe of which our Earth is but an inconſiderable Point. I know you have ſuch an Eſteem and Reverence for any thing that belongs to the Heavens, that I perſwade my ſelf you will read what I have written with ſome Pleaſure: I'm ſure I writ it with a great deal; but as often before, ſo now, I find the Saying of Archytas true, even to the Letter, That tho' a Man were admitted into Heaven to view the wonderful Fabrick of the World,

and the Beauty of the Stars, yet what would otherwiſe be Rapture and Extaſie, would be but a melancholy Amazement if he had not a Friend to communicate it to. I could wiſh indeed that all the World might not be my Judges, but that I might chuſe my Readers, Men like you, not ignorant in Aſtronomy and true Philoſophy; for with ſuch I might promiſe my ſelf a favourable hearing, and not need to make an Apology for daring to vent any thing new to the World. But becauſe I am aware what weak Hands it's likely to fall into, and what a ſevere Sentence I may expect from thoſe whoſe Ignorance or Zeal is too great; it may be worth the while to guard my ſelf beforehand againſt the Aſſaults of thoſe ſort of People.

[The Objections of ignorant Cavillers prevented.]

There's one ſort who knowing nothing of Geometry or Mathematicks, will laugh at it as a whimſical and ridiculous Undertaking. It's an incredible Thing to them to talk of meaſuring the Diſtance and Magnitude of the Stars: And for the Motion of the Earth, they count it, if not a falſe, at leaſt a precarious Opinion; and no wonder then if they take what's built upon ſuch a ſlippery Foundation for the Dreams of a fanciful Head and a diſtemper'd Brain. What ſhould we anſwer to theſe Men, but that their Ignorance is the Cauſe of their Diſlike, and that if they had ſtudied theſe things more, and viewed the Works of Nature nicely, they would have fewer Scruples? But few People having had an opportunity of proſecuting theſe Studies, either for want of Parts, Learning or Leiſure, we cannot blame their Ignorance; and if they reſolve to find fault with us for ſpending time in ſuch Matters, becauſe they do not underſtand the Uſe of them, we muſt appeal to properer Judges.

[Theſe Conjectures do not contradict the holy Scriptures.]

The other ſort, When they hear us talk of new Lands, and Animals, and Creatures endued with as much Reaſon as themſelves, will be ready to cry out, that we ſet up our Conjectures againſt the Word of God, and broach Opinions directly oppoſite to Holy Writ. For we do not there read any thing of the Production of ſuch Creatures, no not ſo much as that they exiſt; nay rather we read the quite contrary. For, That only mentions this Earth with its Animals and Plants, and Man the Lord of them: To ſuch Perſons I anſwer, what has been often urged by others before me: That it's evident, God had no deſign to make a particular Enumeration in the Holy Scriptures, of all the Works of his Creation. When therefore it is plain that under the general Name of Stars or Earth at the Creation, are comprehended all the Heavenly Bodies, even the Attendants upon Jupiter and Saturn, why muſt all that Multitude of Beings which the Almighty Creator has been pleaſed to place upon them, be excluded the Privilege, and not ſuffered to have a Share in the Expreſſion? And theſe Men themſelves can't but know in what Senſe it is that all things are ſaid to be made for the Uſe of Man, not certainly for us to look at through a Teleſcope, for that's very abſurd. Since then the greateſt part of God's Creation, that innumerable multitude of Stars, is placed out of the reach of any Man's Eye; and many of them it's likely, of the beſt Glaſſes, ſo that they don't ſeem to belong to us; is it ſuch an unreaſonable Opinion to think, that there are ſome reaſonable Creatures who ſee and admire thoſe glorious Bodies at a nearer diſtance?

[This Enquiry not over curious.]

But perhaps they'll ſay, it does not become us to be ſo curious and inquiſitive in theſe Things which the Supreme Creator ſeems to have kept for his own Knowledge: For ſince he has not been pleaſed to make any farther Diſcovery or Revelation of them, it ſeems little better than preſumption to make any inquiry into that which he has thought fit to hide. But theſe Gentlemen muſt be told, that they take too much upon themſelves when they pretend to appoint how far and no farther Men ſhall go in their Searches, and to ſet bounds to other Mens Induſtry; as if they knew the Marks that God has placed to Knowledge: or as if Men were able to paſs thoſe Marks. If our Forefathers had been at this rate ſcrupulous, we might have been ignorant ſtill of the Magnitude and Figure of the Earth, or that there was ſuch a Place as America. We ſhould not have known that the Moon is inlightned by the Sun's Rays, nor what the Cauſes of the Eclipſes of each of them are, nor a multitude of other Things brought to light by the late Diſcoveries in Aſtronomy. For what can a Man imagine more abſtruſe, or leſs likely to be known, than what is now as clear as the Sun? Whence it follows, that vigorous Induſtry, and piercing Wit were given Men to make Advances in the Search of Nature, and there's no Reaſon to put any Stop to ſuch Enquiries. I muſt acknowledge that what I here intend to treat of is not of that Nature as to admit of a certain Knowledge; I can't pretend to aſſert any thing as poſitively true (for how is it poſſible) but only to advance a probable Gueſs, the Truth of which every one is at his own liberty to examine. If any one therefore ſhall gravely tell me, that I have ſpent my Time idly in a vain and fruitleſs Enquiry after what by my own acknowledgment I can never come to be ſure of; The Anſwer is, that at this rate he would put down all Natural Philoſophy as far as it concerns it ſelf in ſearching into the Nature [Conjectures not uſeleſs, becauſe not certain.] of Things: In ſuch noble and ſublime Studies as theſe, 'tis a Glory to arrive at Probability, and the Search it ſelf rewards the Pains. But there are many degrees of Probable, ſome nearer Truth than others, in the determining of which lies the chief exerciſe [Theſe Studies uſeful to Religion.] of our Judgment. But beſides the Nobleneſs and Pleaſure of the Studies, may not we be ſo bold as to ſay, they are no ſmall help to the Advancement of Wiſdom and Morality? ſo far are they from being of no uſe at all. For here we may mount from this dull Earth, and viewing it from on high, conſider whether Nature has laid out all her Coſt and Finery upon this ſmall Speck of Dirt. So, like Travellers into other diſtant Countries, we ſhall be better able to judge of what's done at home, know how to make a true Eſtimate of, and ſet its own Value upon every Thing. We ſhall be leſs apt to admire what this World calls Great, ſhall nobly deſpiſe thoſe Trifles the generality of Men ſet their Affections on, when we know that there are a multitude of ſuch Earths inhabited and adorned as well as our own. And we ſhall worſhip and reverence that God the Maker of all theſe things; we ſhall admire and adore his Providence and wonderful Wiſdom which is diſplayed and manifeſted all over the Univerſe, to the Confuſion of thoſe who would have the Earth and all things formed by the ſhuffling Concourſe of Atoms, or to be without beginning. But to come to our Purpoſe.

[Copernicus's Syſtem explained.]

And now becauſe the chief Argument for the Proof of what we intend will be taken from the Diſpoſition of the Planets, among which without doubt, the Earth muſt be counted in the Copernican Syſtem, I ſhall here firſt of all draw two Figures. The firſt is a Deſcription of the Orbs the Planets move in, in that order that they are placed round the Sun, drawn as near as can be in their true Proportions, like what you have ſeen in my Clock at home. The ſecond ſhows the Proportions of their Magnitudes in reſpect of one another and of the Sun, which you know is upon that ſame Clock of mine too. In the firſt the middle Point or Center is the Place of the Sun, round which, in an order that every one knows, are the Orbits of Mercury, Venus, the Earth with that of the Moon about it; then thoſe of Mars, Jupiter and Saturn: and about the two laſt the ſmall Circles that their Attendants move in: about Jupiter four, and about Saturn five. Which Circles as well as that of the Moon are drawn larger than their true Proportion would admit, otherwiſe they could not have been ſeen. You may eaſily apprehend the Vaſtneſs of theſe Orbits by this, that the diſtance of the Earth from the Sun is ten or twelve thouſand of the Earth's Diameters. Almoſt all theſe Circles are in the ſame Plane, declining very little from that in which the Earth moves, call'd The Plane of the Ecliptick. This Plane is cut obliquely by the Axis upon which the Earth turns it ſelf round with reſpect to the Sun in 24 Hours, whence ariſe the Succeſſions of Day and Night: The Axis of the Earth always keeping the ſame Inclination to the Ecliptick (except a ſmall Change beſt known to Aſtronomers) while the Earth itſelf is carried in its yearly Courſe round the Sun, cauſes the regular Order of the Seaſons of the Year: as you may ſee in all Aſtronomers Books. Out of which I ſhall tranſcribe hither the Periods of the Revolutions of the Planets, viz. Saturn moves round the Sun in 29 Years, 174 Days, and 5 Hours: Jupiter finiſhes his Courſe in 11 Years, 317 Days, and 15 Hours: Mars his in about 687 Days. Our Year is 365 Days 6 Hours: Venus's 224 Days 18 Hours: and Mercury's 88 Days. This is the now commonly received Syſtem, invented by Copernicus, and very agreeable to that frugal Simplicity Nature ſhows in all [Arguments for the Truth on't.] her Works. If any one is reſolved to find fault with it, let him firſt be ſure he underſtands it. Let him firſt ſee in the Books of Aſtronomers with how much greater Eaſe and Plainneſs all the Motions of the Stars, and Appearances in the Heavens are explained and demonſtrated in this than either in that of Ptolomy or Tycho. Let him conſider that Diſcovery of Kepler, that the Diſtances of the Planets from the Sun, as well of the Earth as the reſt, are in a fix'd certain proportion to the Times they ſpend in their Revolutions. Which Proportion it's ſince obſerved that their Satellites keep round Jupiter and Saturn. Let him examine what a contradictory Motion they are fain to invent for the Solution of the Polar Star's changing its Diſtance from the Pole. For that Star in the end of the little Bear's Tail which now deſcribes ſo ſmall a Circle round the Pole, that it is not above two Degrees and twenty Minutes, was obſerved about 1820 Years ago, in the Time of Hipparchus, to be above 12: and will within a few Ages more be 45 Degrees diſtant from it: and after 25000 Years more will return to the ſame Place it is now in. Now if with them we allow the Heavens to be turned upon their own Axis, at this rate they muſt have a new Axis every Day: a Thing moſt abſurd, and repugnant to the Nature of all Motion. Whereas nothing is eaſier with Copernicus than to give us Satisfaction in this Matter. Then he may impartially weigh thoſe Anſwers that Galilæus, Gaſſendus, Kepler, and others have given to all Objections propoſed, which have ſo ſatisfied all Scruples, that

generally all Aſtronomers now-a-days are brought over to our Side, and allow the Earth its Motion and Place among the Planets. If he cannot be ſatisfied with all this, he is either one whoſe Dulneſs can't comprehend it, or who has his Belief at another Man's Diſpoſal.

In the other Figure you have the Globes of the Planets, and of the Sun, repreſented to your Eyes as placed near one another. Where [The Proportion of the Magnitude of the Planets, in reſpect of one another, and the Sun.] I have obſerved the ſame Proportion, of their Diameters to that of the Sun, that I publiſhed to the World in my Book of The Appearances of Saturn: namely, the Diameter of the Ring round Saturn is to that of the Sun as 11 as to 37; that of Saturn himſelf about as 5 to 37; that of Jupiter as 2 to 11; that of Mars as 1 to 166; of the Earth as 1 to 111; and of Venus as 1 to 84: to which I ſhall now add that of Mercury obſerved by Hevelius in the Year 1661, but calculated by my ſelf, and found to be as 1 to 290.

If you would know the way that we came to this Knowledge of their Magnitudes, by knowing the Proportion of their Diſtances from the Sun, and the Meaſures of their Diameters, you may find it in the Book before-mentioned: And I cannot yet ſee any Reaſon to make an Alteration in thoſe I then ſettled, altho' I will not ſay they are without their Faults. [The Lamellæ more convenient than Micrometers.] For I can't yet be of their Mind, who think the Uſe of Micrometers, as they call them, is beyond that of our Plates, but muſt ſtill think that thoſe thin Plates or Rods of which I there taught the Uſe, not to detract from the due Praiſes of ſo uſeful an Invention, are more convenient than the Micrometers.

In this proportion of the Planets it is worth while to take notice of the prodigious Magnitude of the Sun in compariſon with the four innermoſt, which are far leſs than Jupiter and Saturn. And 'tis remarkable, that the Bodies of the Planets do not increaſe together with their Diſtances from the Sun, but that Venus is much bigger than Mars.

[The Earth juſtly likened to the Planets, and the Planets to it.]

Having thus explained the two Schemes, there's no Body I ſuppoſe but ſees, that in the firſt the Earth is made to be of the ſame ſort with the reſt of the Planets. For the very Poſition of the Circles ſhows it. And that the other Planets are round like it, and like it receive all the Light they have from the Sun, there's no room (ſince the Diſcoveries made by Teleſcopes) to doubt. Another Thing they are like it in is, that they are moved round their own Axis; for ſince 'tis certain that Jupiter and Saturn are, who can doubt it of the others? Again, as the Earth has its Moon moving round it, ſo Jupiter and Saturn have theirs. Now ſince in ſo many Things they thus agree, what can be more probable than that in others they agree too; and that the other Planets are as beautiful and as well ſtock'd with Inhabitants as the Earth? Or what ſhadow of Reaſon can there be why they ſhould not?

If any one ſhould be at the Diſſection of a Dog, and be there ſhewn the Intrails, the Heart, Stomach, Liver, Lungs and Guts, all the Veins, Arteries and Nerves; could ſuch a Man reaſonably doubt whether there were the ſame Contexture and Variety of Parts in a Bullock, Hog, or any other Beaſt, tho' he had never chanc'd to ſee the like opening of them? I don't believe he would. Or were we thoroughly ſatisfy'd in the

Nature of one of the Moons round Jupiter, ſhould not we ſtraight conclude the ſame of the reſt of them? So if we could be aſſur'd in but one Comet, what it was that is the Cauſe of that ſtrange Appearance, ſhould we not make that a Standard to judge of all others by? [Arguments from their Similitude, of no ſmall weight.] 'Tis therefore an Argument of no ſmall Weight that is fetch'd from Relation and Likeneſs; and to reaſon from what we ſee and are ſure of, to what we cannot, is no falſe Logick. This muſt be our Method in this Treatiſe, wherein from the Nature and Circumſtances of that Planet which we ſee before our Eyes, we may gueſs at thoſe that are farther diſtant from us.

[The Planets are ſolid, and not without Gravity.]

And, Firſt, 'tis more than probable that the Bodies of the Planets are ſolid like that of our Earth, and that they don't want what we call Gravity, that Virtue, which like a Loadſtone attracts whatſoever is near the Body to its Center. And that they have ſuch a Quality, their very Figure is a Proof; for their Roundneſs proceeds only from an equal preſſure of all their Parts tending to the ſame Center. Nay more, we are ſo skilful now-a-days, as to be able to tell how much more or leſs the Gravitation in Jupiter or Saturn is than here; of which Diſcovery and its Author you may read my Eſſay of the Cauſes of Gravitation.

But now to carry the Search farther, let us ſee by what Steps we muſt riſe to the attaining ſome knowledge in the deeper Secrets concerning the State and Furniture of theſe new Earths. And, firſt, how likely is it that they may be ſtock'd with Plants [Have Animals and Plants.] and Animals as well as we? I ſuppoſe no Body will deny but that there's ſomewhat more of Contrivance, ſomewhat more wonderful in the Production and Growth of Plants and Animals, than in Lifeleſs Heaps of inanimate Bodies, be they never ſo much larger; as Mountains, Rocks, or Seas are. For the Finger of God, and the Wiſdom of Divine Providence, is in them much more clearly manifeſted than in the other. One of Democritus's or Cartes's Scholars may venture perhaps to give ſome tolerable Explication of the Appearances in Heaven and Earth, allow him but his Atoms and Motion; but when he comes to Plants and Animals, he'll find himſelf non-plus'd, and give you no likely account of their Production. For every Thing in them is ſo exactly adapted to ſome Deſign, every part of them ſo fitted to its proper Uſe, that they manifeſt an Infinite Wiſdom, and exquiſite Knowledge in the Laws of Nature and Geometry, as, to omit thoſe Wonders in Generation, we ſhall by and by ſhow; and make it an Abſurdity even to think of their being thus happily jumbled together by a chance Motion of I don't know what little Particles. Now ſhould we allow the Planets nothing but vaſt Deſerts, lifeleſs and inanimate Stocks and Stones, and deprive them of all thoſe Creatures that more plainly ſpeak their Divine Architect, we ſhould ſink them below the Earth in Beauty and Dignity; a Thing very unreasonable, as I ſaid before.

Well then, we have gain'd the Point thus far, and the Planets may be allowed ſome Creatures capable of moving themſelves, not at all inferior to ours; and theſe are Animals. And if this be allowed, it almoſt neceſſarily follows, that there muſt be Herbs [Not to be imagin'd too unlike ours.] for Food for them. And as for the Growth and Nouriſhment of all theſe, 'tis no doubt the ſame with ours, ſeeing they have the ſame

Sun to warm and enliven them as ours have.

But perhaps ſome Body may ſay, we conclude too faſt. They will not deny indeed but that there may be Plants and Animals on the Surface of the Planets, that deſerve as well to be provided for by their Creator as ours do: but why muſt they be of the ſame Kind with ours: Nature ſeems to love variety in her Works, and may have made them widely different from ours either in their matter or manner of Growth, in their outward Shape, or their inward Contexture; ſhe may have made them ſuch as neither our Underſtanding nor Imagination can conceive. That's the Thing we ſhall now examine, and whether it be not more likely that ſhe has not obſerv'd ſuch a Variety as they talk of. Nature ſeems moſt commonly, and in moſt of her Works, to affect Variety, 'tis true; But they ſhould conſider 'tis not the Buſineſs of Men to pretend to ſettle how great this Difference and Variety muſt be. Nor does it follow, becauſe it may be Infinite, and out of our Comprehenſion and Reach, that therefore Things in reality are ſo. For ſuppoſe God ſhould have pleaſed to have made all Things in the reſt of the Planets juſt as he has here, the Inhabitants of thoſe Places (if there are any ſuch) would admire his Wiſdom and Contrivance no leſs than if they were widely different; ſeeing they can't come to know what's done in the other Planets. Who doubts but that God, if he had pleaſed, might have made the Animals in America and other diſtant Countries nothing like ours? yet we ſee he has not done it. They have indeed ſome difference in their Shape, and 'tis fit they ſhould, to diſtinguiſh the Plants and Animals of thoſe Countries from ours, who live on this ſide the Earth; but even in this Variety there is an Agreement, an exact Correſpondence in Figure and Shape, the ſame ways of Growth, and new Productions, and of continuing their own Kind. Their Animals have Feet and Wings like ours, and like ours have Hearts, Lungs, Guts, and the Parts ſerving to Generation; whereas all theſe Things, as well with them as us, might, if it had pleaſed Infinite Wiſdom, have been order'd a very different Way. 'Tis plain then that Nature has not exhibited that Variety in her Works that ſhe could, and therefore we muſt not allow that Weight to this Argument, as upon the Account of it to make every Thing in the Planets quite different from what is here. 'Tis more probable that all the Difference there is between us and them, ſprings from the greater or leſs diſtance and influence from that Fountain of Heat and Life the Sun; which will cauſe a Difference not ſo much in their Form and Shape, as in their Matter and Contexture.

[Planets have Water.]

And as for the Matter whereof the Plants and Animals there conſiſt, tho' it is impoſſible ever to come to the Knowledge of its Nature, yet this we may venture to aſſert (there being ſcarce any Doubt of it) that their Growth and Nouriſhment proceeds from ſome liquid Principle. For all Philoſophers agree that there can be no other way of Nutrition; ſome of the Chief among them having made Water to be the Original of all Things: For whatſoever's dry and without Moiſture, is without Motion too; and without Motion, it's impoſſible there ſhould be any Increaſe. But the Parts of a Liquid being in continual Motion one with another, and inſinuating and twiſting themſelves into the ſmalleſt Places, are thereby very proper and apt to add not themſelves only, but whatſoever elſe they may bring along with them, to the Increaſe and Growth of Bodies. Thus we ſee that by the Means of Water the Plants grow,

bloſſom, and bear Fruit; and by the Addition of that only, Stones grow together out of Sand. And there's no doubt but that Metals, Cryſtals, and Jewels, have the ſame Method of Production: Tho' in them there has been no opportunity to make the ſame Obſervation, as well by reaſon of their ſlow Advances, as that they are commonly found far from the Places of their Generation; thrown up I ſuppoſe by ſome Earthquakes, or Convulſions. That the Planets are not without Water, is made not improbable by the late Obſervations: For about Jupiter are obſerved ſome Spots of a darker Colour than the reſt of his Body, which by their continual change ſhow themſelves to be Clouds: For the Spots of Jupiter which belong to him, and never remove from him, are quite different from theſe, being ſometimes for a long time not to be ſeen for theſe Clouds; and again, when theſe diſappear, ſhowing themſelves. And at the going off of theſe Clouds, ſome Spots have been taken notice of in him, much brighter than the reſt of his Body, which remained but a little while, and then were hid from our Sight. Theſe Monſieur Caſſini thinks are only the Reflection from the Snow that covers the Tops of the Hills in Jupiter: But I ſhould rather think that it is only the Colour of the Earth, which happens to be free from thoſe Clouds that commonly darken it.

Mars too is found not to be without his dark Spots, by means of which he has been obſerved to turn round his own Axis in 24 Hours and 40 Minutes; the Length of his Day: but whether he has Clouds or no, we have not had the ſame opportunity of obſerving as in Jupiter, as well becauſe even when he is neareſt the Earth, he appears to us much leſs than Jupiter, as that his Light not coming ſo far, is ſo brisk as to be an Impediment to exact Obſervations: And this Reaſon is as much ſtronger in Venus as its Light is. But ſince 'tis certain that the Earth and Jupiter have their Water and Clouds, there is no Reaſon why the other Planets [But not juſt like ours.] ſhould be without them. I can't ſay that they are exactly of the ſame nature with our Water; but that they ſhould be liquid their Uſe requires, as their Beauty does that they ſhould be clear. For this Water of ours, in Jupiter or Saturn, would be frozen up inſtantly by reaſon of the vaſt diſtance of the Sun. Every Planet therefore muſt have its Waters of ſuch a temper, as to be proportioned to its Heat: Jupiter's and Saturn's muſt be of ſuch a Nature as not to be liable to Froſt; and Venus's and Mercury's of ſuch, as not to be eaſily evaporated by the Sun. But in all of them, for a continual ſupply of Moiſture, whatever Water is drawn up by the Heat of the Sun into Vapours, muſt neceſſarily return back again thither. And this it cannot do but in Drops, which are cauſed as well there as with us, by their aſcending into a higher and colder Region of the Air, out of that which, by reaſon of the Reflection of the Rays of the Sun from the Earth, is warmer and more temperate.

Here then we have found in theſe new Worlds Fields warm'd by the kindly Heat of the Sun, and water'd with fruitful Dews and Showers: That there muſt be Plants in them as well for Ornament as Uſe, we have ſhewn juſt now. And what Nouriſhment, what manner of Growth ſhall we allow [Plants grow and are nouriſhed there as they are here.] them? Probably, there can be no better, nay no other, than what we here experience; by having their Roots faſtned into the Earth, and imbibing its nouriſhing Juices by their tender Fibres. And that they may not be only like ſo many bare Heaths, with nothing but creeping Shrubs and Buſhes, we may allow them ſome nobler and

loftier Plants, Trees, or ſomewhat like them: Theſe being the greateſt, and, except Waters, the only Ornament that Nature has beſtowed upon the Earth. For not to ſpeak of thoſe many uſes that are made of their Wood, there's no one that is ignorant either of their Beauty or Pleaſantneſs. Now what way can any one imagine for a continual Production and Succeſſion of theſe Plants, but their bearing Seed? A Method ſo excellent, that it's the only one that Nature has here made uſe of, and ſo wonderful, that it ſeems to be deſigned not for this Earth alone. In fine, there's the ſame reaſon to think that this Method is obſerved in thoſe diſtant Countries, as there was of its being followed in the remote Quarters of this ſame Earth.

[The ſame true of their Animals.]

'Tis much the ſame in Animals as 'tis in Plants, as to their manner of Nouriſhment, and Propagation of their Kind. For ſince all the living Creatures of this Earth, whether Beaſts, Birds, Fiſhes, Worms, or Inſects, univerſally and inviolably follow the ſame conſtant and fix'd Inſtitution of Nature; all feed on Herbs, or Fruits, or the Fleſh of other Animals that fed on them: ſince all Generation is performed by the impregnating of the Eggs, and the Copulation of Male and Female: Why may not the ſame Rule be obſerved in the Planetary Worlds? For 'tis certain that the Herbs and Animals that are there would be loſt, their whole Species deſtroyed without ſome daily new Productions: except there be no ſuch thing there as Misfortune or Accident: except the Plants are not like other humid Bodies, but can bear Heat, Froſt, and Age, without being dry'd up, kill'd or decay'd: except the Animals have Bodies as hard and durable as Marble; which I think are groſs Abſurdities. If we ſhould invent ſome new Way for their coming into the World, and make them drop like Soland Geeſe from Trees, how ridiculous would this be to any one that conſiders the vaſt Difference between Wood and Fleſh? Or ſuppoſe we ſhould have new ones made every Day out of ſome ſuch fruitful Mud as that of Nile, who does not ſee how contrary this is to all that's reaſonable? And that 'tis much more agreeable to the Wiſdom of God, once for all to create of all ſorts of Animals, and diſtribute them all over the Earth in ſuch a wonderful and inconceivable way as he has, than to be continually obliged to new Productions out of the Earth? And what miſerable, what helpleſs Creatures muſt theſe be, when there's no one that by his Duty will be obliged, or by that ſtrange natural fondneſs, which God has wiſely made a neceſſary Argument for all Animals to take care of their own, will be moved to aſſiſt, nurſe or educate them?

As for what I have ſaid concerning their Propagation, I cannot be ſo poſitive; but the other Thing, namely, that they have Plants and Animals, I think I have fully proved, viz. from hence, that otherwiſe they would be inferiour to our Earth. And by the ſame Argument, they muſt have as great a Variety of both as we have. What this is, will be beſt known to him that conſiders the different Ways our Animals make uſe of in moving from one Place to another. Which may be reduc'd, I think, to theſe, either that they walk upon two Feet or Four; or like Inſects, upon Six, nay ſometimes Hundreds; or that they fly in the Air bearing up, and wonderfully ſteering themſelves with their Wings; or creep upon the Ground without Feet; or by a violent Spring in their Bodies, or paddling with their Feet, cut themſelves a Way in the Waters. I don't believe, nor can I conceive, that there ſhould be any other Way than theſe mentioned.

The Animals then in the Planets muſt make uſe of one or more of theſe, like our amphibious Birds, which can ſwim in Water as well as walk on Land, or fly in the Air; or like our Crocodiles and Sea-Horſes, muſt be Mongrels, between Land and Water. There can no other Method be imagined but one of theſe. For where is it poſſible for Animals to live, except upon ſuch a ſolid Body as our Earth, or a fluid one like the Water, or ſtill a more fluid one than that, ſuch as our Air is? The Air I confeſs may be much thicker and heavier than ours, and ſo, without any Diſadvantage to its Tranſparency, be fitter for the volatile Animals. There may alſo be many ſorts of Fluids ranged over one another in Rows as it were. The Sea perhaps may have ſuch a fluid lying on it, which tho' ten times lighter than Water, may be a hundred Times heavier than Air; whoſe utmoſt Extent may not be ſo large as to cover the higher Places of their Earth. But there's no Reaſon to ſuſpect or allow them this, ſince we have no ſuch Thing; and if we did, it would be of no Advantage to them, for that the former Ways of moving would not be hereby at all increas'd: But when we come to meddle with the Shape of theſe Creatures, and conſider the incredible Variety that is even in thoſe of the different parts of this Earth, and that America has ſome which are no where elſe to be found, I muſt then confeſs that I think it beyond the Force of Imagination to arrive at any knowledge in the Matter, or reach to Probability concerning the Figures of theſe Planetary Animals. Altho' conſidering theſe Ways of Motion we e'en now recounted, they may perhaps be no more different from ours than ours (thoſe of ours I mean that are moſt unlike) are from one another.

If a Man were admitted to a Survey of Jupiter or Venus, he would no doubt find as great a Number and Variety as he had at home. Let us then, that we may make as near a Gueſs at, and as reaſonable a Judgment of the Matter as we can, conſider the many Sorts, and the admirable Difference in the Shapes of our own Animals; running [Great Variety of Animals in this Earth.] over ſome of the Chief of them (for 'twould be tedious to ſet about a general Catalogue) that are notoriouſly different from one another, either in the Figure or ſome peculiar Property belonging to them; as they belong to the Land, or the Water, or the Air. Among the Beaſts we may take notice of the great Diſtance between the Horſe, the Elephant, the Lion, the Stag, the Camel, the Hog, the Ape, the Porcupine, the Tortoiſe, the Cameleon: in the Water, of that between the Whale, and the Sea-Calf, the Skait, the Pike, the Eel, the Ink-Fiſh, the Pourcontrel, the Crocodile, the Flying-fiſh, the Cramp-fiſh, the Crab, the Oiſter, and the Purple-Fiſh: and among Birds, of that between the Eagle, the Oſtrich, the Peacock, the Swan, the Owl, and the Bat: and in Inſects, of that between the Ants, the Spider, the Fly, and the Butterfly; and of that Prodigy in their wonderful change from Worms. In this Roll I have paſs'd by the creeping Kind as one Sort, and skip'd over that vaſt Multitude of leſs different Animals that fill the intermediate Spaces. But be they never ſo many, there is no [And no leſs in the Planets.] reaſon to think that the Planets cannot match them. For tho' we in vain gueſs at the Figures of thoſe Creatures, yet we have diſcover'd ſomewhat of their manner of Life in general; and of their Senſes we ſhall ſpeak more by and by.

[The ſame in Plants.]

The more conſiderable Differences in our Plants ought to be thought on, as well as the other. As in Trees, that between the Fir and the Oak, the Palm, the Vine, the Fig, and the Coco-Nut Tree, and that in the Indies, from whoſe Boughs new Roots ſpring, and grow downwards into the Earth. In Herbs, the Difference is notable between Graſs, Poppy, Colewort, Ivy, Pompions, and the Indian Fig with thick Leaves growing up without any Stalk, and Aloe. Between every one of which again there are many leſs differing Plants not taken notice of. Then the different Ways of raiſing them are remarkable, whether from Seeds, or Kernels, or Roots, or by grafting or inoculating them. And yet in all theſe, whether we conſider the Things themſelves, or the Ways of their Production, I make no doubt but that the Planetary Worlds have as wonderful a Variety as we.

But ſtill the main and moſt agreeable Point of the Enquiry is behind, [Rational Animals in the Planets.] which is the placing ſome Spectators in theſe new Diſcoveries, to enjoy theſe Creatures we have planted them with, and to admire their Beauty and Variety. And among all, that have never ſo ſlightly meddled with theſe Matters, I don’t find any that have ſcrupled to allow them their Inhabitants: not Men perhaps like ours, but ſome Creatures or other endued with Reaſon. For all this Furniture and Beauty the Planets are ſtock’d with ſeem to have been made in vain, without any Deſign or End, unleſs there were ſome in them that might at the ſame time enjoy the Fruits, and adore the wiſe Creator of them. But this alone would be no prevailing Argument with me to allow them ſuch Creatures. For what if we ſhould ſay, that God made them for no other Deſign, but that he himſelf might ſee (not as we do ’tis true; but that he that made the Eye ſees, who can doubt?) and delight himſelf in the Contemplation of them? For was not Man himſelf, and all that the whole World contains, made upon this very account? That which makes me of this Opinion, that thoſe Worlds are not without ſuch a Creature endued with Reaſon, is, that otherwiſe our Earth would have too much the Advantage of them, in being the only part of the Univerſe that could boaſt of ſuch a Creature ſo far above, not only Plants and Trees, but all Animals whatſoever: a Creature that has ſomething Divine in him, that knows, and underſtands, and remembers ſuch an innumerable number of Things; that deliberates, weighs and judges of the Truth: A Creature upon whoſe Account, and for whoſe Uſe, whatſoever the Earth brings forth ſeems to be provided. For every Thing here he converts to his own Ends. With the Trees, Stones, and Metals, he builds himſelf Houſes: the Birds and Fiſhes he ſuſtains himſelf with: and the Water and Winds he makes ſubſervient to his Navigation, as he doth the ſweet Smell and glorious Colours of the Flowers to his Delight. What can there be in the Planets that can make up for its Defects in the want of ſo noble an Animal? If we ſhould allow Jupiter a greater Variety of other Creatures, more Trees, Herbs and Metals, all theſe would not advantage or dignify that Planet ſo much as that one Animal doth ours by the admirable Productions of his penetrating Wit. If I am miſtaken in this, I do not know when to truſt my Reaſon, and muſt allow my ſelf to be but a poor Judge in the true Eſtimate of Things.

[Vices of Men no hindrance to their being the Glory of the Planet they inhabit.]

Nor let any one ſay here, that there's ſo much Villany and Wickedneſs in Man that we have thus magnified, that it's a reaſonable Doubt, whether he would not be ſo far from being the Glory and Ornament of the Planet that enjoys his Company, that he would be rather its Shame and Diſgrace. For firſt, the Vices that moſt Men are tainted with, are no hindrance, but that thoſe that follow the Dictates of true Reaſon, and obey the Rules of a rigid Virtue, are ſtill a Beauty and Ornament to the Place that has the Happineſs to harbour them. Beſides, the Vices of Men themſelves are of excellent Uſe, and are not permitted and allowed in the World without wiſe Deſign. For ſince it has ſo pleaſed God to order the Earth, and every Thing in it as we ſee it is (for it's abſurd to ſay it happen'd againſt his Will or Knowledge) we muſt not think that ſo great a Diverſity of Minds were placed in different Men to no End or Purpoſe: but that this mixture of bad Men with Good, and the Conſequents of ſuch a Mixture, as Misfortunes, Wars, Afflictions, Poverty, and the like, were permitted for this very good End, viz. the exerciſing our Wits, and ſharpening our Inventions; by forcing us to provide for our own neceſſary Defence againſt our Enemies. 'Tis to the Fear of Poverty and Miſery that we are beholden for all our Arts, and for that natural Knowledge which was the Product of laborious Induſtry; and which makes us that we cannot but admire the Power and Wiſdom of the Creator, which otherwiſe we might have paſſed by with the ſame indifference as Beaſts. And if Men were to lead their whole Lives in an undiſturbed continual Peace, in no fear of Poverty, no danger of War, I doubt they would live little better than Brutes, without all knowledge or enjoyment of thoſe Advantages that make our Lives paſs on with Pleaſure and Profit. We ſhould want the wonderful Art of Writing, if its great Uſe and neceſſity in Commerce and War had not forced out the Invention. 'Tis to theſe we owe our Art of Sailing, our Art of Sowing, and moſt of thoſe Diſcoveries of which we are Maſters; and almoſt all the Secrets in experimental Knowledge. So that thoſe very Things on account of which the Faculty of Reaſon ſeems to have been accuſed, are no ſmall helps to its Advancement and Perfection. For thoſe Virtues themſelves, Fortitude and Conſtancy, would be of no uſe if there were no Dangers, no Adverſity, no Afflictions for their Exerciſe and Trial.

If we ſhould therefore imagine in the Planets ſome ſuch reaſonable Creature as Man is, adorn'd with the ſame Virtues, and liable to the ſame Vices, it would be ſo far from degrading or vilifying them, that while they want ſuch a one, I muſt think them inferior to our Earth.

[Reaſon they are not different from what 'tis here.]

But if we allow theſe Planetary Inhabitants ſome ſort of Reaſon, muſt it needs, may ſome ſay, be the ſame with ours? Certainly it muſt, whether we conſider it as applied to Juſtice and Morality, or exerciſed in the Principles and Foundations of Science. For Reaſon with us is that which gives us a true Senſe of Juſtice and Honeſty, Praiſe, Kindneſs and Gratitude: 'tis That that teaches us to diſtinguiſh univerſally between Good and Bad; and renders us capable of Knowledge and Experience in it. And can there be any where any other Sort of Reaſon than this? or can what we call juſt and generous, in Jupiter or Mars be thought unjuſt Villany? This is not at all, I don't ſay probable, but poſſible. For the Aim and Deſign of the Creator is every where the

Preſervation and Safety of his Creatures. Now when ſuch Reaſon as we are Maſters of, is neceſſary for the preſervation of Life, and promoting of Society (a thing that they are not without, as we ſhall ſhow) would it not be ſtrange that the Planetary Inhabitants ſhould have ſuch a perverſe Sort of Reaſon given them, as would neceſſarily deſtroy and confound what it was deſign'd to maintain and defend? But allowing Morality and Paſſions with thoſe diſtant Inhabitants to be ſomewhat different from ours, and ſuppoſing they may act by other Principles in what belongs to Friendſhip and Anger, Hatred, Honeſty, Modeſty, and Comelineſs, yet ſtill there would be no doubt, but that in the Search after Truth, in judging of the Conſequences of Things, in Reaſoning, particularly in that Sort which belongs to Magnitude or Quantity, about which their Geometry (if they have ſuch a Thing) is employ'd, there would be no doubt, I ſay, but that their Reaſon here muſt be exactly the ſame, and go the ſame way to work with ours, and that what's true in one part will hold true over the whole Univerſe; ſo that all the difference muſt lie in the Degrees of Knowledge, which will be proportional to the Genius and Capacity of the Inhabitants.

[They have Senſes.]

But I perceive I am got ſomewhat too far: Let us firſt enquire a little concerning the bodily Senſes of theſe Planetary Perſons; for without ſuch, neither will Life be any Pleaſure to them, nor Reaſon of any Uſe. And I think it very probable, that all their Animals, as well their Beaſts as rational Creatures, are like ours in all that relates to the Senſes: For without the Power of Seeing we ſhould find it impoſſible for Animals to provide Food for themſelves, or be ſore-warn'd of any approaching Danger, ſo as to guard themſelves from it. So that where-ever we plant any Animals, except we wou'd have them lead the Life of Worms or Moles, we muſt allow them Sight; than which nothing can conduce more either to the Preſervation or Pleaſure of their Lives. Then if we conſider the wonderful Nature of Light, and the [Sight.] amazing Artifice in the fit framing the Eye for the Reception of it, we cannot but ſee that Bodies ſo vaſtly remote could not be perceived by us in their proper Figures and juſt Diſtances, any other way than by Sight. For this Senſe, and all others that we know of, muſt proceed from an external Motion. Which in the ſenſe of Seeing muſt come either from the Sun, the fix'd Stars, or Fire: whoſe Particles being put into a very quick Motion, communicate it to the Celeſtial Matter about, whence 'tis convey'd in a very ſhort time to the moſt diſtant parts, juſt like Sound through the Air. If it were not for this Motion of the intermediate Ætherial Matter, we ſhould be all in Darkneſs, and have Sight neither of Sun nor Stars, nor any thing elſe, for all other Light muſt come to us by Reflection from them. This Motion perceived by the Eyes is called Light. And the nice Curioſity of this Perception is admirable, in that it is cauſed by the ſmalleſt Particles of the luminous Body brought to us by that fine Matter, which at the ſame time determine the Coaſt from whence the Motion comes; and in that all theſe different Roads of Motion, theſe Waves croſſing and interfering with one another, are yet no hindrance to every one's free Paſſage. All theſe Things are ſo wiſely, ſo wonderfully contrived, that it's above the Power of humane Wit, to invent or frame any thing like them; nay, it is very difficult ſo much as to imagine and comprehend them. For what can be more amazing, than that one ſmall Part of the Body ſhould be ſo deviſed and framed, as by its means to ſhow us the Shape, the Poſition, the

Diſtance, and all the Motions, nay, and all the Colours, of a Body that is far remote from us, that it may appear the more diſtinct? And then the artful Compoſition of the Eye, drawing an exact Picture of the Objects without it, upon the concave Side of the Choroides, is even above all Admiration, nor is there any Thing in which God has more plainly manifeſted his excellent Geometry. And theſe Things are not only contriv'd and fram'd with ſo great Wiſdom and Skill, as not to admit of better, but to any one that conſiders them attentively, they ſeem to be of ſuch a Nature as not to allow any other Method. For it's impoſſible that Light ſhould repreſent Objects to us at ſo vaſt a diſtance, except by ſuch an intervening Motion; and it's as impoſſible that any other Compoſition of the Eye ſhould be equally fitted to the Reception of ſuch Impreſſions. So that I cannot but think them greatly miſtaken, that maintain theſe Things might have been contrived many other Ways. It's likely then, and credible, that in theſe Things the Planets have an exact correſpondence with us, and that their Animals have the ſame Organs, and uſe the ſame way of Sight that we do. They muſt have Eyes therefore, and two at leaſt we muſt grant them, otherwiſe they would not perceive thoſe Things cloſe to them, nor hardly be able to walk about with Safety. And if we muſt allow them to all Animals for the Preſervation of their Life, how much more muſt they that make more, and more noble Uſes of them, not be deprived of the Bleſſing of ſo advantageous Members? For by them we view the various Flowers, and the elegant Features of Beauty: with them we read, we write, we contemplate the Heavens and Stars, and meaſure their Diſtances, Magnitudes, and Journeys: which how far they are common to the Inhabitants of thoſe Worlds with us, I ſhall preſently examine. But firſt I ſhall enquire whether now we have given them one, we ought alſo to give them the other [Hearing.] four Senſes. And indeed as to Hearing many Arguments perſwade me to give it a Share in the Animals of thoſe new Worlds. For 'tis of great conſequence in defending us from ſudden Accidents; and, eſpecially when Seeing is of no uſe to us, it ſupplies its Place, and gives us ſeaſonable warning of any imminent Danger. Beſides, we ſee many Animals call their Fellow to them with their Voice, which Language may have more in it than we are aware of, tho' we don't underſtand it. But if we do but conſider the vaſt Uſes and neceſſary Occaſions of Speaking on the one ſide, and Hearing on the other, among thoſe Creatures that make uſe of their Reaſon, it will ſcarce ſeem credible that two ſuch uſeful, ſuch excellent Things were deſigned only for us. For how is it poſſible but that they that are without theſe, muſt be without many other Neceſſaries and Conveniences of Life? Or what can they have to recompenſe this Want? Then, if we go ſtill farther, and do but meditate upon the neat and frugal Contrivance of Nature in making the ſame Air, by the drawing in of which we live, by whoſe Motion we ſail, and by whoſe Means Birds fly, for a Conveyance of Sound to our Ears; and this Sound for the Conveyance of another Man's Thoughts to our Minds: Can we ever imagine that ſhe has left thoſe other Worlds deſtitute of ſo vaſt [A Medium to convey Sound to the Ears.] Advantages? That they don't want the Means of them is certain, for their having Clouds in Jupiter puts it paſt doubt that they have Air too; that being moſtly formed of the Particles of Water flying about, as the Clouds are of them gathered into ſmall Drops. And another Proof of it is, the neceſſity of breathing for the preſervation of Life, a Thing that ſeems to be as univerſal a Dictate of Nature, as feeding upon the Fruits of the Earth.

[Touch.]

As for Feeling, it ſeems to be given upon neceſſity to all Creatures that are cover'd with a fine and ſenſible Skin, as a Caution againſt coming too near thoſe Things that may injure or incommode them: and without it they would be liable to continual Wounds, Blows and Bruiſes. Nature ſeems to have been ſo ſenſible of this, that ſhe has not left the leaſt place free from ſuch a Perception. Therefore it's probable that the Inhabitants of thoſe Worlds are not without ſo neceſſary a Defence, and ſo fit a Preſervative againſt Dangers and Miſhaps.

[Smell and Taſte.]

And who is there that doth not ſee the inevitable neceſſity for all Creatures that live by feeding to have both Taſte and Smell, that they may diſtinguiſh thoſe Things that are good and nouriſhing, from thoſe that are miſchievous and harmful? If therefore we allow the Planetary Creatures to feed upon Herbs, Seeds, or Fleſh, we muſt allow them Taſte and Smell, that they may chuſe or refuſe any Thing according as they find it likely to be advantageous or noxious to them.

I know that it hath been a Queſtion with many, whether there might not have been more Senſes than theſe five. [Their Senſes not very different from ours.] If we ſhould allow this, it might nevertheleſs be reaſonably doubted, whether the Senſes of the Planetary Inhabitants are much different from ours. I muſt confeſs, I cannot deny but there might poſſibly have been more Senſes; but when I conſider the Uſes of thoſe we have, I cannot think but they would have been ſuperfluous. The Eye was made to diſcern near and remote Objects, the Ear to give us notice of what our Eyes could not, either in the Dark or behind our Back: Then what neither the Eye nor the Ear could, the Noſe was made (which in Dogs is wonderfully nice) to warn us of. And if any thing eſcapes the notice of the other four Senſes, we have Feeling to inform us of the too near Approaches of it before it can do us any miſchief. Thus has Nature ſo plentifully, ſo perfectly provided for the neceſſary preſervation of her Creatures here, that I think ſhe can give nothing more to thoſe there, but what will be needleſs and ſuperfluous. Yet the Senſes were not wholly deſigned for uſe: but Men from all, and all other Animals from ſome of them, reap Pleaſure as well as Profit, as from the Taſte in delicious Meats; from the Smell in Flowers and Perfumes; from the Sight in the Contemplation of beauteous Shapes and Colours; from the Hearing in the Sweetneſs and Harmony of Sounds; from the Feeling in Copulation, unleſs you pleaſe to count that for a particular Senſe by it ſelf. [They have Pleaſure ariſing from the Senſes.] Since it is thus, I think 'tis but reaſonable to allow the Inhabitants of the Planets theſe ſame Advantages that we have from them. For upon this Conſideration only, how much happier and eaſier a Man's Life is rendred by the enjoyment of them, we muſt be obliged to grant them theſe Bleſſings, except we would engroſs every thing that is good to our ſelves, as if we were worthier and more deſerving than any elſe. But moreover, that Pleaſure which we perceive in Eating or in Copulation, ſeems to be a neceſſary and provident Command of Nature, whereby it tacitly compels us to the preſervation and continuance of our Life and Kind. It is the ſame in Beaſts. So that both for their Happineſs and Preſervation it's very probable the reſt of the Planets are not without it. Certainly when I conſider all theſe Things, how great, noble, and uſeful

they are; when I conſider what an admirable Providence it is that there's ſuch a Thing as Pleaſure in the World, I can't but think that our Earth, the ſmalleſt part almoſt of the Univerſe, was never deſign'd to monopolize ſo great a Bleſſing. And thus much for thoſe Pleaſures which affect our bodily Senſes, but have little or no relation to our Reaſon and Mind. But there are other Pleaſures which Men enjoy, which their Soul only and Reaſon can reliſh: Some airy and brisk, others grave and ſolid, and yet nevertheleſs Pleaſures, as ariſing from the Satisfaction which we feel in Knowledge and Inventions, and Searches after Truth, of which whether the Planetary Inhabitants are not partakers, we ſhall have an opportunity of enquiring by and by. There

There are ſome other things to be conſider'd firſt, in which it's probable they have ſome relation to us. That the Planets have thoſe Elements of Earth, Air and Water, as well as we, I have already made not unlikely. Let us now ſee whether they may not have Fire alſo: which is not ſo properly call'd an Element, as a very quick Motion of the Particles in the inflammable [All the Planets have Fire.] Body. But be it what it will, there are many Arguments for their not being without it. For this Earth is not ſo truly call'd the Place of Fire as the Sun: and as by the Heat of that all Plants and Animals here thrive and live; ſo, no doubt, it is in the other Planets. Since then Fire is cauſed by a moſt intenſe and vigorous Heat, it follows that the Planets, eſpecially thoſe nearer the Fountain of it, have their proportionate degrees of Heat and Fire. And ſince there are ſo many ways of its Production, as by the collection of the Rays of the Sun, by the reflection of Mirrors, by the ſtriking of Flint and Steel, by the rubbing of Wood, by the cloſe loading of moiſt Graſs, by Lightning, by the eruptions of Mountains and Volcanos, it's ſtrange if neither Art ſhould have produced it, nor Nature effected it there by one of theſe many means. Then how uſeful and neceſſary is it to us? By it we drive away Cold, and ſupply the want of the Sun in thoſe Countries where his oblique Rays make a leſs vigorous Impreſſion, and ſo keep a great part of the Earth from being an uninhabited Deſert: which is equally neceſſary in all the Planets, whether we allow them Succeſſion of Seaſons, or a perpetual Spring and Æquinox: for even then the Countries near the Pole would receive but little Advantage from the Heat of the Sun. By the help of this we turn the Night into Day, and thereby make a conſiderable addition to the ſhortneſs of our Lives. Upon all theſe Accounts we ought not to think this Earth of ours enjoy it all alone, and exclude all the other Planets from ſo advantageous and ſo profitable a Gift.

But perhaps it may be asked as well concerning Brutes as rational Creatures, and of their Plants and Trees too, whether they are proportionably [The bigneſs of their Creatures not rightly gueſt at by the bigneſs of the Planets.] larger or leſs than ours. For if the Magnitude of the Planets was to be the Standard of their meaſure, there would be Animals in Jupiter ten or fifteen times larger than Elephants, and as much longer than our Whales, and then their Men muſt be all Giants in reſpect to us. Now tho' I don't ſee any ſo great Abſurdity in this as to make it impoſſible, yet there is no reaſon to think it is really ſo, ſeeing Nature has not always ty'd her ſelf to thoſe Rules which we have thought more convenient for her: For example, the Magnitude of the Planets is not anſwerable to their diſtances from the Sun; but Mars, tho' more remote, is far leſs than Venus: and Jupiter turns round his Axis in ten Hours, when the Earth which is much leſs than him, ſpends 24. But ſince Nature, perhaps ſome will

ſay, has not obſerved ſuch a Regularity in the proportion of Things, for ought we know there may be only a Race of Pygmies about the Bigneſs of Frogs and Mice, poſſeſs'd of the Planets. But I ſhall ſhow that this is very improbable by and by.

[In the Planets are ſorts of rational Creatures as well as here.]

There may ariſe another Queſtion, whether there be in the Planets but one ſort of rational Creatures, or if there be not ſeveral ſorts poſſeſſed of different degrees of Reaſon and Senſe. There is ſomething not unlike this to be obſerved among us. For to paſs by thoſe who have human Shape (altho' ſome of them would very well bear that Enquiry too) if we do but conſider ſome ſorts of Beaſts, as the Dog, the Ape, the Beaver, the Elephant, nay ſome Birds and Bees, what Senſe and Underſtanding they are maſters of, we ſhall be forced to allow, that Man is not the only rational Animal. For we diſcover ſomewhat in them of Reaſon independent on, and prior to all Teaching and Practice.

But ſtill no Body can doubt, but that the Underſtanding and Reaſon of Man is to be preferr'd to theirs, as being comprehenſive of innumerable Things, indued with an infinite memory of what's paſt, and capable of providing againſt what's to come. That there is ſome ſuch Species of rational Creatures in the other Planets, which is the Head and Sovereign of the reſt, is very reaſonable to believe: for otherwiſe, were many Species endued with the ſame Wiſdom and Cunning, we ſhould have them always doing Miſchief, always quarrelling and fighting one with another for Empire and Sovereignty, a Thing that we feel too much of where we have but one ſuch Species. But to let that paſs, our next Enquiry ſhall be concerning thoſe Animals in the Planets which are furniſhed with the greateſt Reaſon, whether it's poſſible to know wherein they employ it, and whether they have made as great Advances in Arts and Knowledge as we in our Planet. Which deſerves moſt to be conſidered and examined of any thing belonging to their Nature; and for the better Performance of it we muſt take our Riſe ſomewhat higher, and nicely view the Lives and Studies of Men.

And in thoſe things wherein Men provide and take care only of what's abſolutely neceſſary for the preſervation of their Life; in defending themſelves from the Injuries of the Air; in ſecuring themſelves againſt the Incurſions of Enemies by Walls; and againſt Fraud and Diſturbances by Laws; in educating their Children, and providing for themſelves and them: In all theſe I can ſee no great reaſon that Man has to boaſt of the Pre-eminency of his Reaſon above Beaſts and other Animals. For moſt of theſe Things they perform with greater Eaſe and Art than we, and ſome of them they have no need of. For that Senſe of Virtue and Juſtice in which Man excels, of Friendſhip, Gratitude and Honeſty, of what uſe are they, but either to put a ſtop to the Wickedneſs of Man, or to ſecure us from mutual Aſſaults and Injuries, Things wherein the Beaſts want no Guide but Nature and Inclination? Then if we ſet before our Eyes the manifold Cares, the Diſturbances of Mind, the reſtleſs Deſires, the dread of Death, that are the reſult of this our Reaſon; and compare them with that eaſy, quiet, and harmleſs Life which other Animals enjoy, we ſhould be apt to wiſh a Change, and conclude that they, eſpecially Birds, lived with more Pleaſure and Happineſs than Man could with all his Wiſdom. For they have as great a Reliſh of bodily Pleaſures as

we, let the new Philoſophers ſay what they will, who would have them to be nothing but Clocks and Engines of Fleſh; a Thing which Beaſts ſo plainly confute by crying and running away from a Stick, and all other Actions, that I wonder how any one could ſubſcribe to ſo abſurd and cruel an Opinion. Nay, I can ſcarce doubt but that Birds feel no ſmall Pleaſure in their eaſy, ſmooth ſailing through the Air; and would much more if they but knew the Advantages it hath above our ſlow and [Men chiefly differ from Beaſts in the Study of Nature.] laborious Progreſſion. What is it then after all that ſets human Reaſon above all other, and makes us preferable to the reſt of the Animal World? Nothing in my Mind ſo much as the Contemplation of the Works of God, and the Study of Nature, and the improving thoſe Sciences which may bring us to ſome knowledge in their Beauty and Variety. For without Knowledge what would be Contemplation? And what difference is there between a Man, who with a careleſs ſupine Negligence views the Beauty and Uſe of the Sun, and the fine golden Furniture of the Heaven, and one who with a learned Niceneſs ſearches into their Courſes; who underſtands wherein the Fix'd Stars, as they are call'd, differ from the Planets, and what is the Reaſon of the regular Viciſſitude of the Seaſons; who by ſound Reaſoning can meaſure the Magnitude and Diſtance of the Sun and Planets? Or between ſuch a one as admires perhaps the nimble Activity and ſtrange Motions of ſome Animals, and one that knows their whole Structure, underſtands the whole Fabrick and Architecture of their Compoſition? If therefore the Principle we before laid down be true, that the other Planets are not inferiour in Dignity to ours, [They have Aſtronomy.] what follows but that they have Creatures not to ſtare and wonder at the Works of Nature only, but who employ their Reaſon in the Examination and Knowledge of them, and have made as great Advances therein as we have? They do not only view the Stars, but they improve the Science of Aſtronomy: nor is there any thing can make us think this improbable, but that fond Conceitedneſs of every Thing that we call our own, and that Pride that is too natural to us to be eaſily laid down. But I know ſome will ſay, we are a little too bold in theſe Aſſertions of the Planets, and that we mounted hither by many Probabilities, one of which, if it chance to be falſe, and contrary to our Suppoſition, would, like a bad Foundation, ruin the whole Building, and make it fall to the Ground. But I would have them to know, that all I have ſaid of their Knowledge in Aſtronomy, has Proofs enough, antecedent to thoſe we now produced. For ſuppoſing the Earth, as we did, one of the Planets of equal Dignity and Honour with the reſt, who would venture to ſay, that no where elſe were to be found any that enjoy'd the glorious Sight of Nature's Theatre? Or if there were any Fellow-Spectators, yet we were the only ones that had dived deep into the Secrets and Knowledge of it? So that here's a Proof not ſo far fetch'd for the Aſtronomy of the Planets, the ſame which we uſed for their having rational Creatures, and enjoying the other Advantages we before talk'd of, which ſerves at the ſame time for the Confirmation of our former Conjectures. But if Amazement and Fear at the Eclipſes of the Moon and Sun gave the firſt occaſion to the Study of Aſtronomy, as probably they did, then it's almoſt impoſſible that Jupiter and Saturn ſhould be without it; the Argument being of much greater force in them, by reaſon of the daily Eclipſes of their Moons, and the frequent ones of the Sun to their Inhabitants. So that if a Perſon diſintereſted in his Judgment, and equally ignorant of the Affairs of all the Planets, were to give his Opinion in this Matter, I don't doubt he would give the Cauſe

for Aſtronomy to thoſe two Planets rather than us.

This Suppoſition of their Knowledge and Uſe of Aſtronomy in the Planetary Worlds, will afford us many new Conjectures about their manner of Life, and their State as to other things.

[And all its ſubſervient Arts.]

For, Firſt: No Obſervations of the Stars that are neceſſary to the Knowledge of their Motions, can be made without Inſtruments; nor can theſe be made without Metal, Wood, or ſome ſuch ſolid Body. Here's a neceſſity of allowing them the Carpenters Tools, the Saw, the Ax, the Plane, the Mallet, the File: and the making of theſe requires the Uſe of Iron, or ſome equally hard Metal. [Geometry and Arithmetick:] Again, theſe Inſtruments can't be without a Circle divided into equal Parts, or a ſtrait Line into unequal. Here's a neceſſity for introducing Geometry and Arithmetick. Then the Neceſſity [And Writing.] in ſuch Obſervations of marking down the Epochas or Accounts of Time, and of tranſmitting them to Poſterity, will force us to grant them the Art of Writing; perhaps very different from ours which is commonly uſed, but I dare affirm not more ingenious or eaſy. For how much more ready and expeditious is our Way, than by that multitude of Characters uſed in China; and how vaſtly preferable to Knots tied in Cords, or the Pictures in uſe among the barbarous People of Mexico and Peru? There's no Nation in the World but has ſome way or other of writing or marking down their Thoughts: So that it's no wonder if the Planetary Inhabitants have been taught it by that great School-miſtreſs Neceſſity, and apply it to the Study of Aſtronomy and other Sciences. In Aſtronomical Matters the Neceſſity of it is moreover apparent from hence, that the Motion of the Stars is as 'twere to be fancied and gueſs'd at in different Syſtems, and theſe Syſtems to be continually improved and corrected, as later and more exact Obſervations ſhall convince the old ones of Faults: all which can never be deliver'd down to ſucceeding Generations, unleſs we make uſe of Letters and Figures.

But after all theſe large and liberal Allowances to Them, they will ſtill be behind-hand with us. For we have ſo certain a Knowledge of the true [And Opticks.] Syſtem and Frame of the Univerſe; we have ſo admirable an Invention of Teleſcopes to help our failing Eye-ſight in the view of the Bigneſs and different Forms of the Planetary Bodies, in the diſcovery of the Mountains, and the Shadows of them on the Surface of the Moon, in the bringing to light an innumerable multitude of Stars otherwiſe inviſible, that we muſt neceſſarily be far their Maſters in that Knowledge. Hence it is almoſt neceſſary (except we have a Mind to flatter and complement our ſelves as the only People that have the Advantage of ſuch excellent Inventions) either to allow the Planetary Inhabitants ſuch ſharp Eyes as not to need them, or elſe the uſe of Glaſſes to help the Deficiency of their Sight. And yet I dare not aſſert this, leſt any one ſhould be ſo diſturbed at the Extravagancy of ſuch an Opinion, as to take the meaſure of my other Conjectures by it, and hiſs them all off, upon the account of this alone.

[Theſe Sciences not contrary to Nature.]

But ſome Body may perhaps object, and that not without reaſon at firſt ſight, that the Planetary Inhabitants it's likely are deſtitute of all refined Knowledge, juſt as the Americans were before they had Commerce with the Europeans. For if one conſiders the Ignorance of thoſe Nations, and of others in Aſia and Africa equally barbarous, it will appear as if the main Deſign of the Creator in placing Men upon the Earth was that they might live, and, in a juſt ſenſe of all the Bleſſings and Pleaſure they enjoy, worſhip the Fountain of their Happineſs; but that ſome few went beyond the Bounds of Nature in their Enquiries after Knowledge. There does not want an Anſwer to theſe Men. For God could not but foreſee the Advances Men would make, in their enquiring into the Heavenly Bodies: that they would diſcover Arts uſeful and advantageous to Life: that they would croſs the Seas, and dig up the Bowels of the Earth. Nothing of all this could happen contrary to the Mind and Knowledge of the Infinite Author of all Things. And if he foreſaw theſe Things would be, he ſo appointed and deſtin'd them to humane kind. And the Studies of Arts and Sciences cannot be ſaid to be contrary to Nature, ſince in the ſearch thereof they are employ'd: eſpecially if we conſider how great the natural, deſire and love of Knowledge, rooted in all Men is. For it's impoſſible this ſhould have been given them upon no Deſign or Account. But they will urge, that if ſuch a Knowledge is natural, if we were born for it, why are there ſo very few, eſpecially in Aſtronomy, that proſecute theſe Studies? For Europe is the only Quarter of the Earth in which there have been any Advancements made in Aſtronomy. And as for the Judicial Aſtrology, which pretends to foretel what is to come, it is ſuch a wretched and oftentimes miſchievous piece of Madneſs, that I do not think it ought to be ſo much as named here. And even in Europe, not one in a hundred Thouſand meddles with theſe Studies. Beſides, its Original and Riſe is ſo late, that many Ages were paſt before the very firſt Rudiments of Aſtronomy or Geometry (which is neceſſary to the learning of it) were known. For every Body is acquainted almoſt with its firſt Beginnings in Egypt and Greece. Add to this, that 'tis not yet above fourſcore Years ſince the bungling Epicycles were diſcarded, and the true and eaſy plain Motion of the Planets was diſcovered. For the Satisfaction of theſe Scruples, to what we ſaid before, concerning the Fore-knowledge of God, may be added this; That God never deſigned we ſhould come into the World Aſtronomers or Philoſophers; theſe Arts are not infuſed into us at our Birth, but were ordered, in long Tracts of Time, by degrees to be the Rewards and Reſult of laborious Diligence; eſpecially thoſe Sciences which are now in debate, are ſo much the more difficult and abſtruſe, that their late Invention and ſlow Progreſs are ſo far from being a Wonder, that it is rather ſtrange they were ever diſcover'd at all. There are but few, I acknowledge one or two perhaps in an Age, that purſue them, or think them their Buſineſs: but their Number will be very conſiderable if we take in thoſe that have lived in all the Ages in which Aſtronomy hath flouriſhed: and no Body can deny them that Happineſs and Contentment which they have pretended to above all others. In fine, it was ſufficient that ſo ſmall a Number ſhould make it their Study, ſo that the Profit and Advantage of their Inventions might but ſpread it ſelf over all the World. Since then the Inhabitants of this Earth, let them be never ſo few, have had Parts and Genius ſufficient for the Attainment of this Knowledge; and there's no reaſon to think the Planetary Inhabitants leſs ingenious or happy than our ſelves; we have gain'd our Point, and 'tis probable that they are as skilful Aſtronomers as we can pretend to be. So that now we

may venture to deduce ſome Conſequences from ſuch a Suppoſition.

We have before ſhow'd the neceſſary Dependence and Connexion, not only of Geometry and Arithmetick, but of Mechanical Arts and Inſtruments with this Science. This leads us naturally to the Enquiry how they can uſe theſe Inſtruments and Engines for the Obſervation of the Stars, how they can write down ſuch their Obſervations, and perform other Things which we do with our Hands. So that we muſt neceſſarily give them [They have Hands.] Hands, or ſome other Member, as convenient for all those Uſes, inſtead of them. One of the ancient Philoſophers laid ſuch Streſs upon the Uſe and Conveniency of the Hands, that he made no ſcruple to affirm, they were the Cauſe and Foundation of all our Knowledge. By which, I ſuppoſe, he meant no more, than that without their Help and Aſſiſtance Men could never arrive to the Improvement of their Minds in natural Knowledge: And indeed not without Reaſon. For ſuppoſe inſtead of them they had had Hoofs like Horſes or Bullocks given them, they might have laid indeed the Model and Deſign of Cities and Houſes in their Head, but they would never have been able to have built them. They would have had no Subject of Diſcourſe but what belong'd to their Victuals, Marriages, or Self-preſervation. They would have been void of all Knowledge and Memory, and indeed would have been but one degree diſtant from brute Beaſts. What could we invent or imagine that could be ſo exactly accommodated to all the deſign'd Uſes as the Hands are? Elephants can lay hold of, or throw any thing with their Proboſcis, can take up even the ſmalleſt Things from the Ground, and can perform ſuch ſurpriſing Things with it, that it has not very improperly been call'd their Hand, tho' indeed it is nothing but a Noſe ſomewhat longer than ordinary. Nor do Birds ſhow leſs Art and Deſign in the Uſe of their Bills in the picking up their Meat, and the wonderful Compoſure of their Neſts. But all this is nothing to thoſe Conveniences the Hand is ſo admirably ſuited to; nothing to that amazing Contrivance in its Capacity of being ſtretched, or contracted, or turned to any Part as Occaſion ſhall require. And then, to paſs by that nice Senſe that the Ends of the Fingers are endued with, even to the feeling and diſtinguiſhing moſt ſorts of Bodies in the Dark, what Wiſdom and Art is ſhow'd in the Diſpoſition of the Thumb and Fingers, ſo as to take up or keep faſt hold of any Thing we pleaſe? Either then the Planetary Inhabitants muſt have Hands, or ſomewhat equally convenient, which it is not eaſy to conceive; or elſe we muſt ſay that Nature has been kinder not only to us, but even to Squirrels and Monkeys than them.

[And Feet.]

That they have Feet alſo ſcarce any one can doubt, that does but conſider what we ſaid but juſt now of Animals different Ways of going along, which it's hard to imagine can be perform'd any other ways than what we there recounted. And of all thoſe, there's none can agree ſo well with the ſtate of the Planetary Inhabitants, as that that we here make uſe of. Except (what is not very probable, if they live in Society, as I ſhall ſhow they do) they have found out the Art of flying in ſome of thoſe Worlds.

[That they are upright.]

The Stature and Shape of Men here does ſhow forth the Divine Providence ſo much in its being ſo fitly adapted to its deſign'd Uſes, that it is not without reaſon that all

the Philoſophers have taken notice of it, nor without Probability that the Planetary Inhabitants have their Eyes and Countenance upright, like us, for the more convenient and eaſy Contemplation and Obſervations of the Stars. For if the Wiſdom of the Creator is ſo obſervable, ſo Praiſe-worthy in the Poſition of the other Members; in the convenient Situation of the Eyes, as Watches in the higher Region of the Body; in the removing of the more uncomely Parts out of ſight as 'twere, we cannot but think he has almoſt obſerved the ſame Method in the Bodies of thoſe remote Inhabitants. Nor [It follows not therefore that they have the ſame Shape with us.] does it follow from hence that they muſt be of the ſame Shape with us. For there is ſuch an infinite poſſible variety of Figures to be imagined, that both the Structure of their whole Bodies, and every part of them, both outſide and inſide, may be quite different from ours. How warmly and conveniently are ſome Creatures cloth'd with Wool, and how finely are others decked and adorn'd with Feathers? Perhaps among the rational Creatures in the Planets there may ſome ſuch diſtinction be obſerv'd in their Garb and Covering; a Thing in which Beaſts ſeem to excel Men in here. Unleſs perhaps Men are born naked, for this reaſon to put them upon employing and exerciſing their Wits, in the inventing and making that Attire that Nature had made neceſſary for them. And 'tis this Neceſſity that has been the greateſt, if not only occaſion of all the Trade and Commerce, of all the Mechanical Inventions and Diſcoveries that we are Maſters of. Beſides, Nature might have another great Conveniency in her Eye, by bringing Men into the World naked, namely, that they might accommodate themſelves to all places of the World, and go thicker or thinner cloth'd, according as the Seaſon and Climate they liv'd in requir'd. There may ſtill be conceived a greater difference between us and the Inhabitants of the Planets; for there are ſome ſort of Animals, ſuch as Oyſters, Lobſters, and Crab-fiſh, whoſe Fleſh is on the inſide of their Bones as 'twere. But that which hinders me from aſcribing ſuch a kind of Frame and Compoſition to the Planetary Inhabitants, is that Nature ſeems to have done it only in a few of the meaneſt Sort of Creatures, and that hereby they would be deprived of that quick eaſy motion of their Hands and Fingers, which is ſo uſeful and neceſſary to them, otherwiſe I ſhould not be much affected with the odd Shape and Figure.

[A rational Soul may inhabit another Shape than ours.]

For 'tis a very ridiculous Opinion, that the common People have got, that 'tis impoſſible a rational Soul ſhould dwell in any other Shape than ours. And yet as ſilly as 'tis, it has been the occaſion of many Philoſophers allowing the Gods no other Shape; nay, the Foundation of a Sect among the Chriſtians, that from hence have the Name of Anthropomorphites. This can proceed from nothing but the Weakneſs, Ignorance, and Prejudice of Men; the ſame as that other concerning humane Shape, that it is the handſomeſt and moſt excellent of all others, when indeed it's nothing but a being accuſtomed to that Figure that makes us think ſo, and a Conceit that we and all other Animals naturally have, that no Shape or Colour can be ſo good as our own. Yet ſo powerful are theſe, that were we to meet with a Creature of a much different Shape from Man, with Reaſon and Speech, we ſhould be much ſurpriſed and ſhocked at the Sight. For if we try to imagine or paint a Creature like a Man in every Thing elſe, but that has a Neck four times as long, and great round Eyes five or ſix times as

big, and farther diſtant, we cannot look upon't without the utmoſt Averſion, altho' at the ſame time we can give no account of our Diſlike.

[The Planetarians not leſs than we.]

When I juſt now mentioned the Stature of the Planetary Inhabitants, I hinted that 'twas improbable they ſhould be leſs than we are. For it's likely, that as our Bodies are made in ſuch a proportion to our Earth, as to render us capable of travelling about it, and making Obſervations upon its Bulk and Figure, the ſame Order is obſerv'd in the Inhabitants of the other Planets, unleſs in this Particular alſo, which is very conſiderable, we would prefer our ſelves to all others. Then ſeeing we have before allow'd them Aſtronomy and Obſervations, we muſt give them Bodies and Strength ſufficient for the ruling their Inſtruments, and the erecting their Tubes and Engines. And for this the larger they are the better. For if we ſhould ſuppoſe them Dwarfs not above the Bigneſs of Rats or Mice, they could neither make ſuch Obſervations as are requiſite; nor ſuch Inſtruments as are neceſſary to thoſe Obſervations. Therefore we muſt ſuppoſe them larger than, or at leaſt equal to, our ſelves, eſpecially in Jupiter and Saturn, which are ſo vaſtly bigger than the Planet which we inhabit.

[They live in Society.]

Aſtronomy, we ſaid before, could never ſubſiſt without the writing down the Obſervations: Nor could the Art of Writing (any more than the Arts of Carpenters and Founders) ever be found out except in a Society of reaſonable Creatures, where the Neceſſities of Life forc'd them upon Invention: So that it follows from hence, (as was before ſaid) that the Planetary Inhabitants muſt in this be like us, that they maintain a Society and Fellowſhip with, and afford mutual Aſſiſtances and Helps to one another. Hereupon we muſt allow them a ſettled, not a wandring Scythian way of living, as more convenient for Men in ſuch Circumſtances. But what follows from hence? Muſt they not have every thing elſe proper for ſuch a manner of living granted them too? Muſt they not have their Governours, Houſes, Cities, Trade and Bartering? Why ſhould they not, when even the barbarous People of America and other Places were at their firſt Diſcovery found to have ſomewhat of that nature in uſe among them. I don't ſay, that Things muſt be the ſame there as they are here. We have many that may very well be ſpared among rational Creatures, and were deſign'd only for the preſervation of Society from all Injury, and for the curbing of thoſe Men who make an ill uſe of their Reaſon to the Detriment of others. Perhaps in the Planets they have ſuch plenty and affluence of all good Things, as they neither need or deſire to ſteal from one another; perhaps they may be ſo juſt and good as to be at perpetual Peace, and never to lie in wait for, or take away the Life of their Neighbour: perhaps they may not know what Anger or Hatred are; and if ſo, they muſt be much happier than we. But it's more likely they have ſuch a mixture of Good with Bad, of Wiſe with Fools, of War with Peace, and want not that School-miſtreſs of Arts Poverty. For, as was before ſhown, ſome good uſe may be made of theſe things, but if not, there is no Reaſon why we ſhould prefer their Condition to our own.

[They enjoy the Pleaſures of Society.]

What I am now going to ſay may ſeem ſomewhat more bold, and yet is not leſs likely than the former. For if theſe Nations in the Planets live in Society, as I have pretty well ſhow'd they do, 'tis ſomewhat more than probable that they enjoy not only the Profit, but the Pleaſures ariſing from Society: ſuch as Converſation, Amours, Jeſting, and Shews. Otherwiſe we ſhould make them live without Diverſion or Merriment; we ſhould deprive them of the great Sweetneſs of Life, which it can't well be without, and give our ſelves ſuch an Advantage over them as Reaſon will by no means admit of.

But to proceed to a farther Enquiry into their Buſineſs and Employment, let's conſider what we have not yet mention'd, wherein they may bear any Likeneſs to us. And firſt we have good Reaſon to believe they build themſelves Houſes, becauſe we are ſure they are not without their Showers. For in Jupiter have been obſerved Clouds, big no doubt with Vapours and Water, which hath been proved by many other Arguments, not to be wanting in that Planet. They have Rain then, for otherwiſe how could all the Vapours drawn up by the Heat of the Sun be diſpoſed of? And Winds, for they are cauſed only by Vapours diſſolved by Heat, and it's plain that they blow in Jupiter by the continual Motion and Variety of the [They have Houſes to ſecure 'em from Weather.] Clouds about him. To protect themſelves from theſe, and that they may paſs their Nights in Quiet and Safety, they muſt build themſelves Tents or Huts, or live in Holes of the Earth. But why may we not ſuppoſe the Planetary Inhabitants to be as good Architects, have as noble Houſes, and as ſtately Palaces as our ſelves? Unleſs we think that every Thing which belongs to our ſelves is the moſt beautiful and perfect that can be. And who are we, but a few that live in a little Corner of the World, upon a Ball ten Thouſand times leſs than Jupiter or Saturn? And yet we muſt be the only skilful People at Building; and all others muſt be our Inferiours in the Knowledge of uniform Symetry; and not be able to raiſe Towers and Pyramids as high, magnificent, and beautiful, as our ſelves. For my part, I ſee no reaſon why they may not be as great Maſters as we are, and have the Uſe of all thoſe Arts ſubſervient to it, as Stone-cutting and Brick-making, and whatſoever elſe is neceſſary for it, as Iron, Lead and Glaſs; or ornamental to it, as Gilding and Picture.

If their Globe is divided like ours, into Sea and Land, as it's evident it is (elſe whence could all thoſe Vapours in Jupiter proceed?) we have great Reaſon to allow them the Art of Navigation, and not vainly ingroſs ſo great, ſo uſeful a Thing to our ſelves. Eſpecially conſidering the great Advantages Jupiter and Saturn have for Sailing, in having ſo many Moons to direct their Courſe, by whoſe Guidance they may attain eaſily to the Knowledge that we are not Maſters of, of the Longitude of Places. And what a Multitude of other Things follow from this Allowance? If they have Ships, they muſt have Sails and Anchors, Ropes, Pullies, and Rudders, which are of particular Uſe in directing a Ship's Courſe againſt the Wind, and in ſailing different Ways with the ſame Gale. And perhaps they may not be without the Uſe of the Compaſs too, for the magnetical Matter, which continually paſſes thro' the Pores of our Earth, is of ſuch a Nature, that it's very probable the Planets have ſomething like [They have Navigation, and all Arts ſubſervient.] it. But there's no doubt but that they muſt have the Mechanical Arts and Aſtronomy, without which Navigation can no more ſubſiſt, than they can without Geometry.

But Geometry ſtands in no need of being prov'd after this manner. Nor doth it want Aſſiſtance from other Arts which depend upon it, but we may have a nearer and ſhorter Aſſurance of their not being without it in thoſe Earths. For that Science is of ſuch ſingular Worth and Dignity, ſo peculiarly imploys the Underſtanding, and gives it ſuch a full Comprehenſion and infallible [As Geometry.] certainty of Truth, as no other Knowledge can pretend to: it is moreover of ſuch a Nature, that its Principles and Foundations muſt be ſo immutably the ſame in all Times and Places, that we cannot without Injuſtice pretend to monopolize it, and rob the reſt of the Univerſe of ſuch an incomparable Study. Nay Nature it ſelf invites us to be Geometricians; it preſents us with Geometrical Figures, with Circles and Squares, with Triangles, Polygones, and Spheres, and propoſes them as it were to our Conſideration and Study, which abſtracting from its Uſefulneſs, is moſt delightful and raviſhing. Who can read Euclid, or Apollonius, about the Circle, without Admiration? Or Archimedes of the Surface of the Sphere, and Quadrature of the Parabola without Amazement? or conſider the late ingenious Diſcoveries of the Moderns with Boldneſs and Unconcernedneſs? And all theſe Truths are as naked and open, and depend upon the ſame plain Principles and Axioms in Jupiter and Saturn as here, which makes it not improbable that there are in the Planets ſome who partake with us in theſe delightful and pleaſant Studies. But what's the greateſt Argument with me, that there are ſuch, is their Uſe, I had almoſt ſaid Neceſſity, in moſt Affairs of humane Life. Now we are got thus far, what if we ſhould venture ſomewhat farther, and ſay, that they have our Inventions of the Tables of Sines, of Logarithms, and Algebra? I know it would ſound very odd, and perhaps a little ridiculous, and yet there's no reaſon but the thinking our ſelves better than all the World, to hinder them from being as happy in their Diſcoveries, and as ingenious in their Inventions as we our ſelves are.

[They have Muſick.]

It's the ſame with Muſick as with Geometry, it's every where immutably the ſame, and always will be ſo. For all Harmony conſiſts in Concord, and Concord is all the World over fix'd according to the ſame invariable Meaſure and Proportion. So that in all Nations the Difference and Diſtance of Notes is the ſame, whether they be in a continued gradual Progreſſion, or the Voice makes skips over one to the next. Nay very credible Authors report, that there's a ſort of Bird in America, that can plainly ſing in order ſix muſical Notes: Whence it follows, that the Laws of Muſick are unchangeably fix'd by Nature, and therefore the ſame Reaſon holds for their Muſick, as we e'en now ſhewed for their Geometry. For why, ſuppoſing other Nations and Creatures, endued with Reaſon and Senſe as well as we, ſhould not they reap the Pleaſures ariſing from theſe Senſes as well as we too? I don't know what Effect this Argument, from the immutable Nature of theſe Arts, may have upon the Minds of others; I think it no inconſiderable or contemptible one, but of as great Strength as that which I made uſe of above to prove that the Planetary Inhabitants had the Senſe of Seeing.

But if they take delight in Harmony, there is no doubt but that they have invented Muſical Inſtruments. For they could ſcarce help lighting upon ſome or other by chance; the Sound of a tight String, the Noiſe of the Winds, or the whiſtling of Reeds,

might have given them the hint. From theſe ſmall Beginnings they perhaps, as well as we, have advanced by degrees to the Uſe of the Lute, Harp, Flute, and many ſtring'd Inſtruments. But altho' the Tones are certain and determinate, yet we find among different Nations a quite different manner and rule for Singing; as formerly among the Dorians, Phrygians, and Lydians, and in our Time among the French, Italians, and Perſians. In like manner it may ſo happen, that the Muſick of the Inhabitants of the Planets may widely differ from all theſe, and yet be very good. But why we ſhould look upon their Muſick to be worſe than ours, there's no reaſon can be given; neither can we well preſume that they want the Uſe of Halſ-Notes and Quarter-Notes, ſeeing the Invention of Halſ-Notes is ſo obvious, and the Uſe of them ſo agreeable to Nature. Nay, to go a Step farther, what if they ſhould excel us in the Theory and practick part of Muſick, and outdo us in Conſorts of vocal and inſtrumental Muſick, ſo artificially compos'd, that they ſhew their Skill by the Mixtures of Diſcords and Concords? and of this laſt ſort 'tis very likely the 5th and 3d are in uſe with them.

This is a very bold Aſſertion, but it may be true for ought we know, and the Inhabitants of the Planets may poſſibly have a greater inſight into the Theory of Muſick than has yet been diſcover'd among us. For if you ask any of our Muſicians, why two or more perfect Fifths cannot be uſed regularly in Compoſition; ſome ſay 'tis to avoid that Sweetneſs and Luſhiouſneſs which ariſes from the Repetition of this pleaſing Chord. Others ſay, this muſt be avoided for the ſake of that Variety of Chords that are requiſite to make a good Compoſition; and theſe Reaſons are brought by Cartes and others. But an Inhabitant of Jupiter or Venus will perhaps give you a better Reaſon for this, viz. becauſe when you paſs from one perfect Fifth to another, there is ſuch a Change made as immediately alters your Key, you are got into a new Key before the Ear is prepared for it, and the more perfect Chords you uſe of the ſame kind in Conſecution, by ſo much the more you offend the Ear by theſe abrupt Changes.

Again, one of theſe Inhabitants perhaps can ſhow how it comes about, that in a Song of one or more Parts, the Key cannot be kept ſo well in the ſame agreeable Tenour, unleſs the intermediate Cloſes and Intervals be ſo temper'd, as to vary from their uſual Proportions, and thereby to bear a little this way or that, in order to regulate the Scale. And why this Temperature is beſt in the Syſtem of the Strings, when out of the Fifth the fourth Part of a Comma is uſually cut off; This ſame thing I have formerly ſhew'd at large.

But for the regulating the Tone of the Voice (as I before hinted) that may admit of a more eaſy proof, and we ſhall give you an Eſſay of it, ſince I have mentioned a thing that is not mere Imagination only: I ſay therefore, if any Perſon ſtrike thoſe Sounds which the Muſicians diſtinguiſh by theſe Letters, C, F, D, G, C, by theſe agreeable Intervals, altogether perfect, interchangeable, aſcending and deſcending with the Voice: Now this latter ſound C will be one Comma, or very ſmall portion lower than the firſt ſounding of C. Becauſe of theſe perfect Intervals, which are as 4 to 3, 5 to 6, 4 to 3, 2 to 3, an account is made in ſuch a Proportion, as 160 to 162. that is, as 80 to 81, which is what they call a Comma. So that if the ſame Sound ſhould be repeated nine times, the Voice would fall near the Matter a greater Tone, whoſe proportion is

as 8 to 9. But this the Senſe of the Ears by no means endures, but remembers the firſt Tone, and returns to it again. Therefore we are compell'd to uſe an occult Temperament, and to ſing theſe imperfect Intervals, from doing which leſs Offence ariſes. And for the moſt part, all Singing wants this Temperament, as may be collected by the aforeſaid Computations. And theſe things we have offer'd to thoſe that have ſome Knowledge in Geometry.

We have ſpoke of theſe Arts and Inventions, which it is very probable the Inhabitants of the Planets partake of in common with us, beſides which it ſeems requiſite to take in many other Things that ſerve either for the Uſe or Pleaſure of their Lives. But what theſe Things are we ſhall the better account for, by laying before us many of thoſe Things which are found among us. I have before mention'd the Variety of Animals and Vegetables, which very much differ from each other, among which there are ſome that differ but little; and I have ſaid, that there are no leſs differences in theſe Things in the Planetary Worlds.

I ſhall now take a ſhort view of the Benefits we receive both from thoſe Herbs and Animals, and ſee whether we may not with very good reaſon conclude that the Planetary Inhabitants reap as great and as many from thoſe that their Countries afford them.

And here it may be worth our while to take a Review of the Variety and Multitude of our Riches. For Trees and Herbs do not only ſerve us for Food, they in their delicious Fruits, theſe in their Seeds, Leaves and Roots; but Herbs moreover furniſh us with Phyſick, and Trees with Timber for our Houſes and Ships. Flax, by the means of thoſe two uſeful Arts of Spinning and Weaving, affords us Clothing. Of Hemp or Matweed we twiſt our ſelves Thread and ſmall Ropes, the former of which we employ in Sails and Nets, the latter in making larger Ropes for Maſts and Anchors. With the ſweet Smells and [The Advantages we reap from Herbs and Animals.] beauteous Colours of Flowers we feaſt our Senſes: and even thoſe of them, that offend our Noſtrils, or are miſchievous to our Bodies, are ſeldom without excellent Uſes: or were made perhaps by Nature as a Foil to ſet off, and make us the more value the Good by comparing them with theſe. What vaſt Advantages and Profit do we reap from the Animals? The Sheep give us Clothing, and the Cows afford us Milk: and both of them their Fleſh for our Suſtenance. Aſſes, Camels, and Horſes do, what if we wanted them we muſt do our ſelves, carry our Burdens; and the laſt of them we make uſe of, either themſelves to carry us, or in our Coaches to draw us. In which we have ſo excellent, ſo uſeful an Invention of Wheels, that I can't ſuppoſe the Planets to enjoy Society and all its Conſequences, and be without them. Whether they are Pythagoreans there, or feed upon Fleſh as we do, I dare not affirm any Thing. Tho' it ſeems to be allowed Men to feed upon whatſoever may afford them Nouriſhment, either on Land, or in Water, upon Herbs, and Pomes, Milk, Eggs, Honey, Fiſh, and no leſs upon the Fleſh of many Birds and Beaſts. But it is a ſurpriſing thing! that a rational Creature ſhould live upon the Ruin and Deſtruction of ſuch a number of other his Fellow-Creatures! And yet it does not ſeem at all unnatural, ſince not only he, but even Lions, Wolves, and other ravenous Beaſts, prey upon Flocks of other harmleſs Things, and make mere Fodder of them; as Eagles do of Pidgeons and Hares; and large Fiſh of the helpleſs

little ones. We have different ſorts of Dogs for Hunting, and what our own Legs cannot, that their Noſe and Legs can help us to. But the Uſe and Profit of Herbs and Animals are not the only Things they are good for, but they raiſe our Delight and Admiration when we conſider their various Forms and Natures, and enquire into all their different ways of Generation: Things ſo infinitely multifarious, and ſo delightfully amazing, that the Books of natural Philoſophers are deſervedly filled with their Encomiums. For even in the very Inſects, who can but admire the ſix-corner'd Cells of the Bees, or the artificial Web of a Spider, or the fine Bag of a Silk-worm, which laſt affords us, with the Help of incredible Induſtry, even Shiploads of ſoft delicate Clothing. This is a ſhort Summary of thoſe many profitable Advantages the animal and herbal World ſerve us with.

But this is not all. The Bowels of the Earth likewiſe contribute much to Man's Happineſs. For what Art and Cunning does he employ in finding, in digging, in trying Metals, and in melting, refining, and tempering them? [And from Metals.] What Skill and Nicety in beating, drawing or diſſolving Gold, ſo as with inconſiderable Changes to make every Thing he pleaſes put on that noble Luſtre? Of how many and admirable Uſes is Iron? and how ignorant in all Mechanical Knowledge were thoſe Nations that were not acquainted with it, ſo as to have no other Arms but Bows, Clubs, and Spears, made of Wood. There's one Thing indeed we have, which it's a Queſtion whether it has done more harm or good, and that is Gun-powder made of Nitre and Brimſtone. At firſt indeed it ſeem'd as if we had got a more ſecure Defenſe than former Ages againſt all Aſſaults, and could eaſily guard our Towns, by the wonderful Strength of that Invention, againſt all hoſtile Invaſions: but now we find it has rather encouraged them, and at the ſame time been no ſmall Occaſion of the Decay of Valour, by rendring it and Strength almoſt uſeleſs in War. Had the Grecian Emperor who ſaid, Virtue was ruin'd only when Slings and Rams firſt came into uſe, liv'd in our Days, he might well have complain'd; eſpecially of Bombs, againſt which neither Art nor Nature is of ſufficient Proof: but which lays every Thing, Caſtles and Towers, be they never ſo ſtrong, even with the Ground. If for nothing elſe, yet upon this one account, I think we had better have been without the Diſcovery. Yet, when we were talking of our Diſcoveries, it was not to be paſs'd over, for the Planets too may have their miſchievous as well as uſeful Inventions.

We are happier in the Uſes for which the Air and Water ſerves us; both of which helps us in our Navigation, and furniſhes us with a Strength ſufficient, without any Labour of our own, to turn round our Mills and Engines; Things which are of uſe to us in ſo many different Employments. For with them we grind our Corn, and ſqueeze out our Oil; with them we cut Wood, and mill Cloth, and with them we beat our Stuff for Paper. An incomparable Invention! Where the naſtieſt uſeleſs Scraps of Linen are made to produce fine white Sheets. To theſe we may add the late diſcovery of Printing, which not only preſerves from Death Arts and Knowledge, but makes them much eaſier to be attained than before. Nor muſt we forget the Arts of Engraving and Painting, which from mean Beginnings have improved to that Excellence, that nothing that ever ſprung from the Wit of Man can claim Pre-eminence to them. Nor is the way of melting and blowing Glaſſes, and of poliſhing and ſpreading Quick-ſilver over Looking-Glaſſes, unworthy of being mentioned, nor above all, the admirable uſes

that Glaſſes have been put to in natural Knowledge, ſince the Invention of the Teleſcope and Microſcope. And no leſs nice and fine is the Art of making Clocks, ſome of which are ſo ſmall as to be no weight to the Bearer; and others ſo exact as to meaſure out the Time in as ſmall Portions as any one can deſire: the Improvement of both which the World owes to my [1]Inventions.

[From the diſcoveries of our Age.]

I might add much here of the late Diſcoveries, moſt of them of this Age, which have been made in all ſorts of Natural Knowledge as well as in Geometry and Aſtronomy, as of the Weight and Spring of the Air, of the Chymical Experiments that have ſhown us a way of making Liquors that ſhall ſhine in the Dark, and with gentle moving ſhall burn of themſelves. I might mention the Circulation of the Blood through the Veins and Arteries, which was underſtood indeed before; but now, by the help of the Microſcope, has an ocular demonſtration in the Tails of ſome Fiſhes: of the Generation of Animals, which now is found to be performed no otherwiſe than by the Seed of one of the ſame kind; and that in the Seed of the Male are diſcover'd, by the help of Glaſſes, Millions of ſprightly little Animals, which it's probable are the very Offspring of the Animals themſelves: a ſurpriſing thing, and never before now known!

[The Planets have, tho' not theſe ſame, yet as uſeful Inventions.]

Thus have I put together all theſe late Diſcoveries of our Earth: and now, tho' perhaps ſome of them may be common to the Planetary Inhabitants with us, yet that they ſhould have all of them is not credible. But then they may have ſomewhat to make up that Defect, others as good and as uſeful, and as wonderful, that we want. We have allow'd that they may have rational Creatures among them, and Geometricians, and Muſicians: We have prov'd that they live in Societies, have Hands and Feet, are guarded with Houſes and Walls: Wherefore if a Man could be carried thither by ſome powerful Genius, ſome Mercury, I don't doubt 'twould be a very curious ſight, curious beyond all Imagination, to ſee the odd ways, and the unuſual manner of their ſetting about any thing, and their ſtrange methods of living. But ſince there's no hopes of our going ſuch a Journey, we muſt be contented with what's in our Power: we may ſuppoſe our ſelves there, and inquire as far as we can into the Aſtronomy of each Planet, and ſee in what manner the Heavens preſent themſelves to their Inhabitants. We ſhall make ſome Obſervations of the Eminence of each of them, in reſpect of their Magnitude, and number of Moons they have to wait on them; and ſhall propoſe a new Method of coming to ſome Knowledge of the incredible diſtance of the ſix'd Stars. But firſt after this long and deep Thoughtfulneſs we will give our ſelves a little Reſt, and ſo put an end to this Book.

New Conjectures concerning the Planetary Worlds.

BOOK the Second.

'Twas a pretty many Years ago that I chanc'd to light upon Athanaſius Kircher's Book, call'd The Ecſtatick Journey, which treats of the nature of the Stars, and of the Things that are to be found in the Superficies of the Planets: I wondered to ſee nothing there

of what I had often thought not improbable, but quite other Things, nothing but a Heap of idle unreaſonable Stuff: which I was the more confirm'd in, when, after the writing of the former part, I ran over the Book again. And I thought mine were very conſiderable and weighty Matters if compar'd with Kircher's. That other People may be ſatisfied in this, and ſee how vainly thoſe, who caſt off the only Foundations of Probability in ſuch Matters, which we have all the way made uſe of, pretend to philoſophize in this caſe, I think it will not be beſide the Purpoſe to beſtow ſome few Reflections upon that Book.

[Kircher's Journey in Ecſtacy examin'd.]

That ingenious Man ſuppoſing himſelf carried by ſome Angel thro' the vaſt Spaces of Heaven, and round the Stars, tells us, he ſaw a great many things, ſome of which he had out of the Books of Aſtronomers, the reſt are the Product of his own Fancy and Thoughts. But, before he enters upon his Journey, he lays down theſe two Things as certain; that no Motion muſt be allowed the Earth, and that God has made nothing in the Planets, no not ſo much as Herbs, which has either Life or Senſe in it. Leaving then the Syſtem of Copernicus, he chuſes Tycho for his Guide. But when he ſuppoſes all the fix'd Stars to be Suns, and round each of them places their Planets, here (againſt his Will I ſuppoſe) he has unawares made an infinite number of Copernican Syſtems. All which, beſide their own Motion, he abſurdly makes to be carried, with an incredible ſwiftneſs, in twenty four Hours round the Earth. Since moſt of theſe Worlds are out of the Reach of any Man's ſight, as he owns they are, I cannot think for what purpoſe he makes ſo many Suns to ſhine upon deſolate Lands (like our Earth in every thing, he ſays, only that they have neither Plants nor Animals) where there's no one to whom they ſhould give light. And from hence he ſtill falls into more and more Abſurdities. And becauſe he could find no other uſe of the Planets, even in our Syſtem, he is forc'd to beg Help of the Aſtrologers; and would have all thoſe vaſt Bodies made upon no other account than that the whole Univerſe might be preſerved and continue ſecure by their means, and that they might govern the Mind of Man by their various and regular Influences. Accordingly, to gratify Aſtrology, he ſays that Venus was the moſt pleaſant Place, every thing fine and handſome, its Light gentle, its Waters ſweet and purling, and it ſelf beſet all about with ſhining Chryſtals. In Jupiter he found wholeſome and ſweet Gales, delicate Waters, and a Land ſhining like Silver. For from theſe two Planets it ſeems, Men have all that is happy and healthful poured down upon them; and all that renders them handſome and lovely, wiſe and grave, is owing to their Influences. Mercury had I don't know what Airineſs and Briskneſs in it; whence Men derive, when they are firſt born, all their Wit and Cunning. Mars was nothing but infernal, ſtinking, black Flames and Smoke: and Saturn was all melancholy, dreadful, naſty, and dark: for theſe are the Planets (I don't know why, but all Fortune-tellers hate them) that bring all the Plagues and Miſchiefs that we feel upon us, and would exerciſe their Spite ſtill more, unleſs they were ſometimes mitigated and corrected by the benign and kind Influences of the other Planets. All this and ſuch like Stuff his Genius teaches him. Which he makes give a ſerious Anſwer to this idle Queſtion, Whether a Jew or Heathen could be duly and rightly baptized in the Waters of Venus? Of him too he learns that the Heaven of the fix'd Stars is not made of ſolid Matter, but of a thin fluid, wherein an innumerable company of Stars

and Suns lie floating here and there, not chain'd down to any Place, (thus far he's in the right) and deſcribing in the Space of a Day theſe prodigious Circles round the Earth. He forgets here, if there were ſuch a Motion, with what an incredible ſwiftneſs they would fly off from every part of their Orbits. But I ſuppoſe the Intelligences that he has plac'd in them are to take care of that, thoſe Angels that preſide over, and regulate their Motions. And in that he follows a company of Doctors that harbour'd that idle fancy of Ariſtotle upon no Account or Conſideration. But Copernicus has freed thoſe Intelligences of all that Labour and Trouble, only by bringing in the Motion of the Earth: which, if upon no other Account, every one that is not blind purpoſely, muſt own to be neceſſary upon this. I dare ſay Kircher, if he had dar'd freely to ſpeak his Mind, could have afforded us better ſort of Things than theſe. But when he could not have that liberty, I think he might as well have let the whole Matter alone. But enough of this; let's have have done with this famous Author: And now that we have ventur'd to place Spectators in the Planets, let us examine each of them, and ſee what their Years, Days, and Aſtronomy are.

[The Syſtem of the Planets in Mercury.]

To begin with the innermoſt and neareſt the Sun: We know that Mercury is three times nearer that vaſt Body of Light than we are. Whence it follows that they ſee him three times bigger, and feel him nine times hotter than we do. Such a degree of Heat would be intolerable to us, and ſet afire all our dry'd Herbs, our Hay and Straw that we uſe. And yet there is no doubt but that the Animals there, are made of ſuch a Temper, as to be but moderately warm, and the Plants ſuch as to be able to endure the Heat. The Inhabitants of Mercury, it's likely, have the ſame opinion of us that we have of Saturn, that we muſt be intolerably cold, and have little or no Light, we are ſo far from the Sun. There's reaſon to doubt, whether the Inhabitants of Mercury, tho' they live ſo much nearer the Sun, the Fountain of Life and Vigour, are much more airy and ingenious than we. For if we may gueſs at them by what we ſee here, we ſhall not be obliged to grant it. The Inhabitants of Africa and Braſil, that have got for their Share the hotteſt Places in the Earth, being neither ſo wiſe nor ſo induſtrious as thoſe that belong to colder and more temperate Climates; they have ſcarce any Arts or Knowledge among them; and thoſe of them that live upon the very Shore, underſtand little or no Navigation. Nor can I be willing to make all that vaſt number that muſt inhabit thoſe two large Planets, Jupiter and Saturn, and have ſuch noble Attendance, mere dull Blockheads, or without as much Wit as our ſelves, tho' they are ſo far more diſtant from the Sun. The Aſtronomy of thoſe that live in Mercury, and the appearance of the Planets to them, oppoſite at certain times to the Sun, may be eaſily conceived by the Scheme of the Copernican Syſtem in the former Part. At the times of theſe Oppoſitions Venus and the Earth muſt needs appear very bright and large to them. For if Venus ſhines ſo glorioufly to us when ſhe is new and horned, ſhe muſt neceſſarily in oppoſition to the Sun, when ſhe is full, be at leaſt ſix or ſeven times larger, and a great deal nearer to the Inhabitants of Mercury, and afford them Light ſo ſtrong and bright, that they have no reaſon to complain of their want of a Moon. What the Length of their Days are, or whether they have different Seaſons in the Year, is not yet diſcovered, becauſe we have not yet been able to obſerve whether his Axis have any inclination to his Orbit, or what Time he ſpends in his diurnal

Revolution about his own Axis. And yet ſeeing Mars, the Earth, Jupiter and Saturn, have certainly ſuch Succeſſions, there's no reaſon to doubt but that he has his Days and Nights as well as they. But his Year is ſcarce the fourth part ſo long as ours.

The Inhabitants of Venus have much the ſame Face of Things as thoſe in Mercury, only they never ſee him in oppoſition to the Sun, which is occaſioned by his never removing above 38 degrees, or thereabouts, from it. The Sun appears to them larger by half in his Diameter, and above twice in his Circumference, than to us: and by conſequence affords them but twice as much Light and Heat, ſo that they are nearer our Temperature than Mercury. Their Year is compleated in ſeven and a half of our Months. In the Night our Earth, when 'tis on the other ſide of the Sun from Venus, muſt needs ſeem much larger and lighter to Venus than ſhe doth ever to us; and then they may eaſily ſee, if their Eyes be not weaker than ours, our conſtant Attendant the Moon. I have often wonder'd that when I have view'd Venus when ſhe is neareſt to the Earth, and reſembled an Half-moon, juſt beginning to have ſomething like Horns, through a Teleſcope of 45 or 60 Foot long, ſhe always appeared to me all over equally lucid, that I can't ſay I obſerved ſo much as one Spot in her, tho' in Jupiter and Mars, which ſeem much leſs to us, they are very plainly perceiv'd. For if Venus had any ſuch Thing as Sea and Land, the former muſt neceſſarily ſhow much more obſcure than the other, as anyone may ſatisfy himſelf, that from a very high Mountain will but look down upon our Earth. I thought that perhaps the too brisk Light of Venus might be the occaſion of this equal appearance; but when I uſed an Eye-glaſs that was ſmok'd for the Purpoſe, it was ſtill the ſame Thing. What then, has Venus no Sea, or do the Waters there reflect the Light more than ours do, or their Land leſs? Or rather (which is moſt probable in my Opinion) is not all that Light we ſee reflected from an Atmoſphere ſurrounding Venus, which being thicker and more ſolid than that in Mars or Jupiter, hinders our ſeeing any thing of the Globe it ſelf, and is at the ſame time capable of ſending back the Rays that it receives from the Sun? For it is certain that if we looked on the Earth from the outſide of the Atmoſphere, we ſhould not perceive ſuch a difference as we do from a Mountain; but by reaſon of the interpoſed Atmoſphere, we ſhould obſerve very little Diſparity between Sea and Land. 'Tis the ſame Thing that hinders us from ſeeing the Spots in the Moon as plain in the Day as in the Night, becauſe the Vapours that ſurround the Earth being then enlighten'd by the Rays of the Sun, are an Impediment to our Proſpect.

[In Mars.]

But Mars, as I ſaid before, has ſome Parts of him darker than other ſome. By the conſtant Returns of which his Nights and Days have been found to be of about the ſame length with ours. But the Inhabitants have no perceivable Difference between Summer and Winter, the Axis of that Planet having very little or no Inclination to his Orbit, as has been diſcover'd by the Motion of his Spots. Our Earth muſt appear to them almoſt as Venus doth to us, and by the Help of a Teleſcope will be found to have its Wane, Increaſe, and Full, like the Moon: and never to remove from the Sun above 48 Degrees, by whoſe Diſcovery they ſee it, as well as Mercury and Venus, ſometimes paſs over the Sun's Disk. They as ſeldom ſee Venus as we do Mercury. I am apt to believe, that the Land in Mars is of a blacker Colour than that of Jupiter or the Moon,

which is the reaſon of his appearing of a Copper Colour, and his reflecting a weaker Light than is proportionable to his diſtance from the Sun. His Body, as I obſerved before, tho' farther from the Sun, is leſs than Venus. Nor has he any Moon to wait upon him, and in that, as well as Mercury and Venus, he muſt be acknowledged inferiour to the Earth. His Light and Heat is twice, and ſometimes three times leſs than ours, to which I ſuppoſe the Conſtitution of his Inhabitants is anſwerable.

[Jupiter and Saturn the moſt eminent of the Planets both for bigneſs and attendants.]

If our Earth can claim pre-eminence of the fore-mention'd Planets, for having a Moon to attend upon it, (for its Magnitude can make but a ſmall difference) how much Superiour muſt Jupiter and Saturn be to thoſe three and the Earth alſo? For whether we conſider their Bulk, in which they far exceed all the others, or the Number of Moons that wait upon them, it's very probable that they are the chief, the primary Planets in our Syſtem, in compariſon with which the other four are nothing, and ſcarce worth mentioning. For the eaſier Conception of their vaſt Diſparity, I have thought fit to add a Scheme of our Earth, with the Moon's Orbit, and the Globe of the Moon itſelf, and the Syſtems of [Fig. 3.] Jupiter and Saturn, where I have drawn every thing as near the true Proportion as poſſible. Jupiter you ſee is adorned with four, and Saturn with five Moons, all placed in their reſpective Orbits. The Moons about Jupiter we owe to Galilæo, 'tis well known: and any one may imagine he was in no ſmall Rapture at the Diſcovery. The outermoſt but one, and brighteſt of Saturn's, it chanc'd to be my lot, with a Teleſcope not above 12 foot long, to have the firſt ſight of in the Year 1655. The reſt we may thank the induſtrious Caſſini for, who uſed the Glaſſes of Joſ. Campanus's grinding, firſt of 36, and afterwards of 136 foot long. He has often, and particularly in the Year 1672, ſhew'd me the Third and Fifth. The Firſt and Second he gave me notice of by Letters in the Year 1684; but they are ſcarce ever to be ſeen, and I can't poſitively ſay, I had ever that Happineſs; but am as ſatisfied that they are there, as if I had; not in the leaſt ſuſpecting the Credit of that worthy Man. Nay, I am afraid there are One or Two more ſtill behind, and not without reaſon. For between the Fourth and Fifth there's a Diſtance not at all proportionable to that between all the others: Here, for ought I know, there may be a Sixth; or perhaps there may be another without the Fifth that may yet have eſcaped us: for we can never ſee the Fifth but in that part of his Orbit, which is towards the Weſt: for which we ſhall give you a very good reaſon.

Perhaps when Saturn comes into the Northern Signs, and is at a good height from the Horizon (for at the writing of this he is at his loweſt) you may happen to make ſome new Diſcoveries, good Brother, if you would but make uſe of your two Teleſcopes of 170 and 210 Foot long; the longeſt, and the beſt I believe now in the World. For tho' we have not yet had an opportunity of obſerving the Heavens with them (as well by reaſon of their Unwieldineſs, as for the Interruption of our Studies by your Abſence) yet I am ſatisfied of their Goodneſs by our trial of them one Night, in reading a Letter at a vaſt diſtance by the Help of a Light. I cannot but think of thoſe times with Pleaſure, and of our diverting Labour in poliſhing and preparing ſuch Glaſſes, in inventing new Methods and Engines, and always puſhing forward to ſtill

greater and greater Things. But to return to the Figures, of which there remains ſomething further to be ſaid.

[The proportion of the Diameter of Jupiter, and of the Orbs of his Satellites, to the Orbit of the Moon round the Earth.]

I have there made the Diameter of Jupiter about two third parts of our diſtance from the Moon: for the Diameter of Jupiter is above twenty times bigger than that of the Earth; which is about a thirtieth part of the Moon's diſtance. The Orbit of the outermoſt of Jupiter's Satellites is to that of the Moon round the Earth, as 8 and ½ is to 1. And each of theſe Moons, by the Shadow they make upon Jupiter, cannot be leſs than our Earth. Their [The Periods of Jupiter's Moons.] Periods, that I may not omit them, are according to Caſſini's Account theſe. That of the inmoſt is one day, 18 hours, 28 minutes, and 36 ſeconds. The Second ſpends 3 days, 13 hours, 13 min., 52 ſeconds in going round him. The Third 7 days, 3 hours, 59 min., 40 ſec. The Fourth 16 days, 18 hours, 5 min., 6 ſec. The Diſtance of the innermoſt from Jupiter himſelf is 2⅝ of his Diameters. That of the Second is 4 and a half: Of the Third 7 and one ſixth part: Of the Fourth 12 and two thirds, of the ſame [And Saturn's.] Diameters. The Innermoſt of Saturn's Satellites moves round him in 1 day, 21 hours, 18 min., 31 ſec. The Second in 2 days, 17 hours, 41 min., 27 ſec. The Third in 4 days, 13 hours, 47 min., 16 ſec. The Fourth in 15 days, 22 hours, 41 min., 11 ſec. The Fifth in 79 days, 7 hours, 53 min., 57 ſec. Their Diſtances from the Center of Saturn are, that of the firſt almoſt one, that is 39 fortieth parts of the Diameter of his Ring; that of the ſecond one and a quarter of thoſe Diameters; of the third one and three quarters of them; of the fourth four, or according to my Calculation, but 3 and a half; of the 5th 12, which were found with vaſt Pains and Labour.

Now can any one look upon, and compare theſe Syſtems together, without being amazed at the vaſt Magnitude and noble Attendance of theſe two Planets, in reſpect of this little Earth of ours? Or can they force themſelves to think, that the wiſe Creator has diſpoſed of all his Animals and Plants here, has furniſh'd and adorn'd this Spot only, and has left all thoſe Worlds bare and deſtitute of Inhabitants, who might adore and worſhip him; or that all thoſe prodigious Bodies were made only to twinkle to, and be ſtudied by ſome few perhaps of us poor Mortals?

[This proportion true according to all modern Obſervations.]

I do not doubt but there will be ſome who will think we are very much miſtaken about the Magnitude of theſe Planets. For will you pretend to make them who are taken up in admiring the Largeneſs of this Globe, its multitude of Nations, Cities, and Empires; can you pretend I ſay to make them ever believe that there are Places in compariſon of which the Earth is as inconſiderable as this Figure would make it? But they ought to be inform'd, that theſe Proportions are thoſe which the beſt Aſtronomers of this Age have agreed upon. For if the Earth be diſtant from the Sun ten or eleven thouſand of its own Diameters, according to the Accounts of Monſieur Caſſini in France, and Mr. Flamſted in England, wherein they made uſe of very exact Obſervations of the Parallaxes of Mars; or if, according to a very probable Conjecture of mine, it be diſtant twelve thouſand, then the Magnitudes of the other Orbs will very

near anſwer the Proportions here ſettled.

[The apparent magnitude of the Sun in Jupiter, and a way of finding what Light they there enjoy.]

But to return to Jupiter. The Sun appears to them who are upon it five times leſs than to us, and conſequently they have but the five and twentieth part of the Light and Heat that we receive from it. But that Light is not ſo weak as we imagine, as is plain by the Brightneſs of that Planet in the Night; and alſo from hence, that when the Sun is ſo far eclipſed to us, as that only the 25th part of his Disk remains uncovered, he is not ſenſibly darken'd. But if you have a mind exactly to know the Quantity of Light that Jupiter enjoys, you may take a Tube of what Length you pleaſe. Let one end of it be cloſed with a Plate of Braſs, or any ſuch thing, in the middle of which there muſt be a Hole, whoſe Breadth muſt have the ſame proportion to the length of the Tube, as the Chord of 6 Minutes bears to the Radius; that is, about as one is to 570. Let the Tube be turned ſo to the Sun, that no Light may fall upon a white Paper placed at the End of it, but what comes through the little Hole at the other end of the Tube. The Rays that come through this will repreſent the Sun upon the Paper of the ſame Brightneſs that the Inhabitants of Jupiter ſee it in a clear Day. And if removing the Paper you place your Eye in the ſame Place, you will ſee the Sun of the ſame Magnitude and Brightneſs as you would were you in Jupiter.

[And in Saturn.]

If you make the Hole twice as little in breadth, you will ſee the ſame in Saturn. And altho' his Light be but the hundredth part of ours, yet you ſee it makes him ſhine tolerably bright in a dark Night. But in both theſe Planets, if there ever be any cloudy Days, it muſt be very dark in compariſon of us; yet without doubt the Inhabitants have no more reaſon to complain of the want of Light, than our Owls and Batts, to whom the Twilight or the Night itſelf is more agreeable than the Brightneſs of the Day.

[In Jupiter their days are five Hours.]

But it's a little ſtrange, that when Jupiter is ſo much bigger than our Planet, their Days and Nights ſhould be but five of our Hours. By this we may ſee that Nature has not obſerv'd that proportion that their Bulk ſeems to require, ſeeing in Mars the Days are very little different from ours. But in the length of their Years, that is, in the Revolution of the Planets round the Sun, there is an exact proportion to their diſtances from the Sun followed. For as the Cubes of their Diſtances, ſo are the Squares of their Revolutions, as Kepler firſt found out. Which proportion the Moons of Jupiter and Saturn keep in their Courſes round thoſe [Always of the ſame length.] Planets. As the Years and Days in Jupiter are different from ours in this reſpect, ſo are the Days in another; namely, that they are all of the ſame length. For they there enjoy a perpetual Equinox, their Axis having little or no inclination to their Orbit, as the Earth's has, as has been diſcovered by Teleſcopes. The Countries that lie near their Poles have little or no Heat, by reaſon the Rays of the Sun fall ſo obliquely upon them; but then they are freed from the Inconveniency that ours are troubled with, of tedious long half-year Nights, and have the conſtant returns of Day and Night every five Hours. Indeed ſuch ſhort Days would not be agreeable to us, but we think our ſelves

much better done by, that ours are more than twice as long, tho’ upon no other account, but that whatever is our own, we are apt to imagine, muſt be beſt.

The reſt of the Planets are ſo near the Sun (Mars himſelf never being above 18 degrees from it) that in Jupiter they have the ſight only of Saturn. But we cannot deny but that their four Moons ſtand them in greater ſtead than our one doth us, if ’twere only that they ſeldom know any ſuch Thing as to be without Moonſhiny Nights. And they are of great Advantage to them, as we ſaid before, in their Navigation, if they have any ſuch thing. Not to mention the pleaſant Sights of their frequent Conjunctions and Eclipſes, Things that they are ſeldom a Day without.

Saturn enjoys all thoſe Pleaſures and Advantages in a ſtill higher Degree, as well for his five Moons, as for the delightful Proſpect that the Ring about him affords his Inhabitants Night and Day. But we will give an account of their Aſtronomy, as we have done of the reſt of the Planets.

[They ſee the fix’d Stars juſt as we do.]

And firſt of all we ſhall obſerve what we might have remark’d before, but which will be more ſtrange here, that the fix’d Stars appear to them of the ſame Figure and Magnitude, and with the ſame degree of Light that they do to us: and this, by reaſon of their immenſe diſtance, of which we ſhall have occaſion to ſpeak by and by. In compariſon with which the Space that a Bullet-ſhot out of a Cannon could travel in 25 Years, would be almoſt nothing.

Their Aſtronomers have all the ſame Signs of the Bear, the Lion, Orion, and the reſt, but not turning upon the ſame Axis with us: for that’s different in all the Planets.

As Jupiter can ſee no Planet but Saturn, ſo Saturn knows of no Planet but Jupiter; which appears to him much as Venus doth to us, never removing above 37 Degrees from the Sun. The Length of their Days I cannot determine: But if from the Diſtance and Period of his innermoſt Attendant, and comparing it with the innermoſt of Jupiter’s, a Man may venture to give a Gueſs, they are very little different from Jupiter’s, 10 Hours or ſomewhat leſs. But whereas in Jupiter theſe are equally divided between Light and Darkneſs, the Inhabitants of Saturn muſt perceive a more ſenſible difference than we, eſpecially between Summer and Winter. For our Axis inclines to the Plane of the Ecliptick but 23 degrees and a half, but there’s above 31. Upon this Account his Moons muſt decline very much from the Path that the Sun ſeems to move in, and his Inhabitants can never have a full Moon but juſt at the Equinoxes; Two of which fall out in 30 of our Years. ’Tis this Poſition of the Axis too that is the Cauſe of thoſe delightful Appearances, and wonderful Proſpects that its Inhabitants enjoy: For the better underſtanding of which I ſhall draw a Figure of Saturn with his Ring about him: in which the Proportion between the Diameters of the Globe and Ring is as 9 to 4. And the empty Space between them is of the ſame Breadth with the Ring itſelf. All Obſervations conſpire to prove that That is of no great Thickneſs, altho’ if we ſhould allow it ſix hundred German Miles, I think, conſidering its Diameter, we ſhould not overdo the Matter.

Suppoſe then, agreeable to what has been ſaid, the Globe of Saturn, [Fig. 4.] whoſe Poles are A, B. G N is the Diameter of the Ring, as you view it ſideways, repreſenting

a narrow Oval. Thoſe that live about the Poles within the Arches C A D, E B F, each of which are 54 Degrees, (if the Cold will ſuffer any Body to live there) never have a Sight of the Ring. [The Appearances of the Ring in Saturn.] From all other parts it is continually to be ſeen for fourteen Years and nine Months, which is juſt half their Year. The other Half it is hid from their View. Thoſe then that dwell between the Polar Circle C D, and the Equator T V, all that time that the Sun enlightens the Part oppoſite to them, have every Night the Sight of a Piece of it H G L, much in the Shape of a ſhining Bow, which comes from the Horizon, but is darken'd in the Middle by the Shadow of Saturn G H, which reaches moſt commonly to the outermoſt Rim of it. But after Midnight that Shadow by little and little begins to move towards the right Hand to thoſe in the Northern, but the Left to thoſe in the Southern Hemiſphere. In the Morning it diſappears, leaving behind it a Likeneſs indeed of a Bow, but much paler and weaker than our Moon is in the Day time. For they, as I ſaid before, have an Atmoſphere, or an Air ſurrounding them enlightened by the Sun. Otherwiſe Night and Day they would have their Ring, their Moons, and all the fix'd Stars, equally conſpicuous. Another thing that muſt make the Sight of their Ring very curious, is, that by ſome Spots in it, it is diſcover'd to turn round upon it ſelf: A thing that thoſe that are ſo near cannot but take notice of, when we that live at this Diſtance can deſcry a great Inequality, the inſide of it being brighter much than the outſide is. When the Shadow of the Globe falls upon that part of the Ring G H, the Shadow of the Ring at the ſame time darkens another Part of the Globe about P F, which otherwiſe would have the Sun upon it. So that there is always a Zone of the Globe P Y F E, ſometimes of a larger extent than at others, which is depriv'd of the Sight both of the Sun and Ring for a conſiderable time, the latter of which hides ſome part of the Stars from it too. And certainly an amazing Thing it muſt be, all of a ſudden to have the Sun intercepted and to become as dark as Midnight, without ſeeing any Cauſe of ſuch an Accident. All which time their Moons are their only Comfort. The other half of the Year the Hemiſphere T B V enjoys the ſame Light that T A U before did, and then this undergoes thoſe long Eclipſes that That before ſuffer'd. At the Equinoxes, when the Sun is in the ſame Plane with the Ring, the Inhabitants of Saturn cannot well perceive it: no not even we with our Glaſſes, by reaſon of its Darkneſs. This happens when Saturn, view'd from the Sun, is advanced one and twenty degrees and a half in Virgo or Piſces, as I have ſhow'd formerly in my Syſtem of Saturn: Where there is an Account given of the Riſings of the Sun above the Ring, throughout all the Saturnian Year.

With Saturn in this Scheme you have the Globes of the Earth and Moon drawn in their true proportion, to put you in mind again of a Thing worth remembring, viz. how very ſmall our Habitation is when compar'd with that Globe or the Ring about it. And now any one, I ſuppoſe, can frame to himſelf a Picture of the Night in Saturn, with two Arches of the Ring, and five Moons ſhining about, and adorning him. This then is what I had to ſay to the primary Planets.

We are now come a little lower, to make an enquiry into the Attendants of theſe Planets, eſpecially our own. And here we ſhall not only conſider their Aſtronomy, but ſhall alſo ſearch into their Furniture and Ornament, if they are found to have any ſuch thing, which we have deferred conſidering till now.

[Very little to be ſaid of the Moon.]

And here one would think that when the Moon is ſo near us, and by the Means of a Teleſcope may be ſo nicely and exactly obſerv'd, it ſhould afford us Matter for more probable Conjectures than any of the other remote Planets. But it is quite otherwiſe, and I can ſcarce find any thing to ſay of it, becauſe I have not a Planet of the ſame Nature before my Eyes, as in all the primary ones I have. For they are of the ſame kind with our Earth; and ſeeing all the Actions, and every thing that is here, we may make a reaſonable Conjecture at what we cannot ſee in thoſe Worlds.

[The Guards of Jupiter and Saturn of the ſame nature with our Moon.]

But this we may venture to ſay, without fear, that all the Attendants of Jupiter and Saturn are of the ſame Nature with our Moon, as going round them, and being carried with them round the Sun juſt as the Moon is with the Earth. Their Likeneſs reaches to other Things too, as you'll ſee by and by. Therefore whatſoever we can with reaſon affirm or conjecture of our Moon (and we may ſay a little of it) muſt be ſuppos'd with very little Alteration to belong to the Satellites of Jupiter and Saturn, as having no reaſon to be at all inferior to that.

[The Moon hath Mountains.]

The Surface of the Moon then is found, by the leaſt Teleſcopes of about three or four Foot, to be diverſified with long Tracts of Mountains, and again with broad Valleys. For in thoſe Parts oppoſite to the Sun you may ſee the Shadows of the Mountains, and often diſcover the little round Valleys between them, with a Hillock or two perhaps riſing out of them. Kepler from the exact roundneſs of them would prove that they are ſome vaſt work of the rational Inhabitants. But I can't be of his mind, both for their incredible Largeneſs, and that they might eaſily be occaſioned by natural Cauſes. Nor can I find any thing like Sea there, tho' he and many others are of the contrary Opinion I know. For thoſe vaſt Countries which appear darker than the other, commonly taken for and called by the Names of Seas, are diſcover'd with a good long Teleſcope, to be full of little round Cavities; whoſe Shadow falling within themſelves, makes them appear of that Colour: and thoſe large Champains there in the Moon you will find not to be always even and ſmooth, if you look carefully upon them: neither of which two [But no Sea.] Things can agree to the Sea. Therefore thoſe Plains in her that ſeem brighter than the other Parts, muſt conſiſt, I ſuppoſe, of a whiter ſort of Matter than they. Nor do I believe [Nor Rivers.] that there are any Rivers, for if there were, they could never eſcape our Sight, eſpecially if they run between the Hills as ours do. Nor have they [Nor Clouds.] any Clouds to furniſh the Rivers with Water: For if they had, we ſhould ſometimes ſee one part of the Moon darkened by them, and ſometimes another, whereas we have always the ſame Proſpect of her.

[Nor Air, and Water.]

'Tis certain moreover, that the Moon has no Air or Atmoſphere ſurrounding it as we have. For then we could never ſee the very outermoſt Rim of the Moon ſo exactly as we do, when any Star goes under it, but its Light would terminate in a gradual faint Shade, and there would be a ſort of a Down as it were about it; not to mention

that the Vapours of our Atmoſphere conſiſt of Water, and conſequently that where there are no Seas or Rivers, there can be no Atmoſphere. This is that notable difference between the Moon and us that hinders all probable Conjectures about it. If we could but once be ſure that there were Seas and Rivers in it, it would be no weak Argument to prove that it has alſo all other Furniture which belongs to our Earth, and the Opinion of Xenophanes might be true, that it has its Inhabitants, Cities, and Mountains. But as ’tis, I cannot imagine how any Plants or Animals, whole whole nouriſhment comes from liquid Bodies, can thrive in a dry, waterleſs, parch’d Soil.

[The Conjecture of its Plants and Animals very dubious.]

What then, is it credible that this great Ball was made for nothing but to give us a little Light in the Night-time, or to raiſe our Tides in the Sea? May there not be ſome People there that may have the Pleaſure of ſeeing our Earth turn upon itſelf, preſenting them ſometimes with a Proſpect of Europe and Africa, and then of Aſia and America; ſometimes half of it bright, and ſometimes full? And muſt all thoſe Moons round Jupiter and Saturn be condemned to the ſame Uſeleſneſs? I do not know what to ſay concerning it, becauſe I know of nothing like them to found a Conjecture upon. And yet ’tis not improbable that thoſe great and noble Bodies have ſomewhat or other growing and living upon them, though very different from what we ſee and enjoy here. Perhaps their Plants and Animals may have another ſort of Nouriſhment there. Perhaps the Moiſture of the Earth there is but juſt ſufficient to cauſe a Miſt or Dew, which may be very ſuitable to the Growth of their Herbs. This I remember is Plutarch’s Opinion, in his Dialogue upon this Subject. For in our Earth a very little Water drawn from the Sea into Dew, and falling down again upon the Herbs, would be ſufficient for all our Needs, without any Rain or Showers. But theſe are mere Gueſſes, or rather Doubts, but yet they are the [Jupiter’s and Saturn’s Moons turn always the ſame Side to them.] beſt we can make of this, and all thoſe other Moons: for, as I ſaid before, they are all of the ſame nature, which is proved likewiſe by this, that as our Moon can afford us the Sight never but of one Side of her, ſo they turn always the ſame Face to their primary Planets. It may perhaps ſeem ſtrange, how we ſhould come to know this; but ’tis no hard matter, after that Obſervation which I juſt now made, that the outermoſt of Saturn’s Moons can never be ſeen but when ſhe is on the Weſt-ſide of her Planet. The reaſon of which is plainly this, that one Side of her is darker, and does not reflect the Light ſo much as the other, which when it is turned towards us, we cannot ſee by reaſon of its weak Light. This always happening when ’tis Eaſt of him, and never on the other Side, is a manifeſt proof that ſhe always keeps the ſame Side toward Saturn. Now ſince the outermoſt of Saturn’s and our Moon carry themſelves thus to the Planets round which they move, who can well doubt it of all the reſt round Jupiter and Saturn? And there’s a very good reaſon for it, namely, that the matter of which thoſe Moons conſiſt, being heavier, and more ſolid on the Side that is averſe from us, than on that which we have the Sight of, does conſequently fly with a greater force from the Centre of its Orbit: for otherwiſe, according to the Laws of Motion, it ſhould turn the ſame Side always, not to its Planets, but to the ſame fix’d Stars.

This Poſition of the Moons, in reſpect of their Planets, muſt occaſion a great many very ſurprizing Appearances to their Inhabitants, if they have any, which is very doubtful, but may for the preſent be ſuppos'd. An enquiry into our Moon may ſerve for all the reſt. Its Globe is divided into two Parts, in ſuch a manner, that thoſe who live on one Side never loſe the ſight of us, and thoſe on the other never enjoy it. Except only ſome few who live on the Confines of each of theſe, who loſe us, and ſee us again by turns. The Earth to them muſt ſeem [The Aſtronomy of the Inhabitants of the Moon.] much larger than the Moon doth to us, as being in Diameter above four times bigger. But that which is moſt ſurprizing, is, that Night and Day they ſee it always in the very ſame part of the Heaven, as if it never moved: ſome of them as if 'twas falling upon their Heads: others ſomewhat above the Horizon, and others always in the Horizon, ſtill turning upon it ſelf, and preſenting them every twenty four Hours with a View of all its Countries, even of thoſe that lie near the Poles (I could wiſh my ſelf in the Moon only for the ſight of them) yet unknown and undiſcovered by us. They have it in its monthly Wane and Increaſe, they ſee it half, and horned, and full, by turns, juſt as we do the Body of the Moon. But the Light that they receive of us is five times larger than what we receive from them. So that in dark Nights that part that hath the Advantage of being towards us, receives a very glorious Light from us, tho' Kepler thought otherwise. Their Days are always of the ſame Length with their Nights; and the Sun riſing and ſetting to them but once in one of our Months, makes the time both of their Light and Darkneſs to be equal to 15 of our Days. If their Bodies were of the ſame Materials with ours, thoſe that have the Sun pretty high in their Horizon, would be almoſt roaſted in ſuch long Days. For the Sun is not farther from them than he is from us. This will be the Caſe of thoſe that live upon the Borders of the two Hemiſpheres we mentioned; but thoſe that live under the Poles of the Moon will be juſt about as hot as our Whale-fiſhers about Iſland and Nova Zemla are, in the Summer-time: who are in ſo little danger of being roaſted, that in the middle of their Summer, in their Days of three Months length, they very often find it extreme Cold. I call thoſe the Poles of the Moon, round which the fix'd Stars ſeem to turn to its Inhabitants, which are different from ours, and alſo from thoſe of the Ecliptick, although they move round theſe latter, at the diſtance of five Degrees, in a period of nineteen Years. Their Year they count by the Motion of the Stars, and their return to the Sun, and 'tis the ſame with ours. They can eaſily do it, becauſe they have the Stars Day and Night, notwithſtanding the Light of the Sun: for they have no Atmoſphere (which is the only reaſon that we don't every Day enjoy the ſame Sight) to hinder their Obſervations. Nor have they any Clouds to obſtruct their View, ſo that it is eaſier for them to find out the Courſes of the Planets, but more difficult to make a true Syſtem of them. For they will be apt to lay a wrong Foundation, by ſuppoſing that their Earth ſtands ſtill, which will lead them into more dangerous Errors than [This may be applied to the Moons about Jupiter and Saturn.] ever it did us. All that I have ſaid belongs as well to Jupiter's and Saturn's Satellites as to our Moon, in reſpect of the Planets they move round. The Length of their Day and Night is always equal to the Time of their Revolution: For example, the fifth Moon moves round Saturn in 80 Days, and the Days and Nights there are equal to Forty of ours. Both their Summer and Winter (Saturn moving round the Sun in thirty Years) are fifteen Years long. Therefore it is impoſſible but that their way of living muſt be very different from ours,

having ſuch tedious Winters, and ſuch long watching and ſleeping times.

Having thus explained the primary and ſecondary Planets round the Sun, we ſhould next ſet about the third Sort, the Sun and fix'd Stars; but before we do that, it would be worth while to ſet before you at once, in a clearer and more plain Method than hitherto, the Magnificence and Fabrick of the Solar Syſtem. Which we can't poſſibly do in ſo ſmall a Space as one of our Leaves will but admit of, becauſe the Bodies of the Planets are ſo prodigiouſly ſmall in companion of their Orbs. But what is wanting in Figure ſhall be made up in Words. Going back then to the firſt Scheme, ſuppoſe another [Fig. 1.] like it, and proportionable, drawn upon a very large ſmooth Plain; whoſe outermoſt Circle repreſenting the Orb of Saturn, muſt be conceived three hundred and ſixty Foot in Semidiameter. In which you muſt place the Globe and Ring of Saturn of that [Fig. 2.] Bigneſs as the 2d Figure ſhows you. Let all the other Planets be ſuppoſed every one in his own Orbit, and in the middle of all the Sun, of the ſame Bigneſs that That Figure repreſents, namely, about four Inches in Diameter. And then the Orbit or Circle in which the Earth moves, which the Aſtronomers call the Magnus Orbis, muſt have about ſix and thirty Foot in Semidiameter. In which the Earth muſt be conceived moving, not bigger than a grain of Millet, and her Companion the Moon ſcarcely perceivable, moving round her in a Circle a little more than two Inches Diameter, as in [Fig. 5.] the Figure here adjoined, where the Line A B repreſents a ſmall portion of that Circle which the Earth moves in: the ſmall Circle therein C is the Earth, and the Circle D E the Path of the Moon round it, in which the Body of the Moon is D.

The outermoſt of Saturn's Moons moves in an Orbit whoſe Semidiameter is 29 Inches; that of Jupiter in a ſomewhat ſmaller, whoſe Semidiameter is 19 and a quarter.

And thus we have a true and exact Deſcription of the Sun's Palace, where the Earth will be Twelve thouſand of its Semidiameters diſtant from him, which in German Miles makes above ſeventeen Millions. But perhaps we may have a clearer Comprehenſion of this vaſt Length, by comparing it with ſome very ſwift Motion after the Example of Heſiod the Poet, who imagin'd that an Anvil let fall from the Top of Heaven, reach'd the Earth the tenth Day of its Journey, and in ten more arriv'd at the Bottom of Hell, the end of it: ſo making the Earth the mid-way between Heaven and Hell. I ſhan't make uſe of the Anvil, but of ſomething as good, namely, a Bullet ſhot out of a great Gun, which may travel perhaps in a Moment, or Pulſe of an Artery, about a hundred Fathom, as is proved by thoſe Experiments that Merſennus in a Treatiſe of his relates; by which the Sound was found to extend itſelf eighty hundredth parts in the [The immenſe diſtance between the Sun and Planets illuſtrated.] ſame time. I ſay then, that ſuppoſing a Bullet to move with this Swiftneſs from the Earth to the Sun, it would ſpend 25 Years in its Paſſage. To make a Journey from Jupiter to the Sun, would require 125, and from Saturn thither 250 Years. This account depends upon the meaſure of the Earth's Diameter, which, according to the accurate Obſervations of the French, is 6538594 times ſix Paris Feet, one Degree being 57060 of that Meaſure. This ſhows us how vaſt thoſe Orbs muſt be, and how inconſiderable this Earth, the Theatre upon which all our mighty Deſigns, all our

Navigations, and all our Wars are tranſacted, is when compared to them. A very fit Conſideration, and Matter of Reflection, for thoſe Kings and Princes who ſacrifice the Lives of ſo many People, only to flatter their Ambition in being Maſters of ſome pitiful Corner of this ſmall Spot. But to return to the matter in hand, now we have given you an account of the Sun's proportion to thoſe Orbs and Bodies, we'll ſee what more we can ſay of him.

[No ground for Conjecture in the Sun.]

And ſome have thought it not improbable but that the Sun himſelf has alſo his Inhabitants. But upon what reaſon I cannot imagine, there being leſs ground for a Probability in him than in the Moon. For we are not yet ſure, whether he be a ſolid or liquid Globe; altho', if my Notion of Light be true, upon that account I ſhould rather think him liquid: which his Roundneſs and equal diſtribution of his Light to all parts are an Argument for. For that very ſmall inequality on his Surface, which is diſcovered by the Teleſcopes, (and that not always neither) which makes Men fancy they ſee boiling Seas and belching Mountains of Fire, is nothing but the trembling Motion of the Vapours our Atmoſphere is full of near the Earth; which is likewiſe [The Faculæ in the Sun not eaſily ſeen.] the Cauſe of the Stars twinkling. Nor could I ever have the Luck to diſcern thoſe bright Spots in the Sun which they boaſt as much of as they do of his dark ones, which latter I have very often ſeen; ſo that I have very good Reaſon to doubt whether there be any thing in the Sun brighter than the Sun itſelf. For by the moſt exact Obſervations, I could never find any ſuch pretended to be ſeen any where but juſt about his dark Spots; and it is no great wonder that thoſe Parts which are ſo near the darker, ſhould appear ſomewhat [By reaſon of its Heat no Inhabitants like ours can live in the Sun.] brighter than the reſt. That the Sun is extremely hot and fiery, is beyond all diſpute, and ſuch Bodies as ours could not live one Moment in ſuch a Furnace. We muſt ſuppoſe a new ſort of Animals then, ſuch as we have no Idea or Likeneſs of among us, ſuch as we can neither imagine nor conceive: which is as much as to ſay, that we can make no Suppoſition at all about them. No doubt that glorious and vaſt Body was made for ſome noble End and Uſe, and fram'd with excellent Deſign. And I think we all very well know and feel its Uſefulneſs in that effuſion of Light and Heat to all the Planets round it; in the Preſervation and Happineſs of all living Creatures, and that not only in our Ball, but in thoſe vaſt Globes of Jupiter and Saturn, not contemptible when compared with its own. Theſe are ſuch great, ſuch wiſe Ends, that it is not ſtrange that the Sun ſhould have been made, if it had been only upon their account. For, as for Kepler's Fancy, that he hath another Office, namely, to help on the Motion of the Planets in their own Orbs, by turning about his own Axis (which he would fain eſtabliſh in his Epitome of the Copernican Syſtem) I ſhall give good Reaſons why I cannot aſſent to it.

[The ſix'd Stars so many Suns.]

Before the Invention of Teleſcopes, it ſeemed to contradict Copernicus's Opinion, to make the Sun one of the fix'd Stars. For the Stars of the firſt Magnitude being eſteem'd to be about three Minutes Diameter; and Copernicus (obſerving that tho' the Earth changed its Place, they always kept the ſame diſtance from us) having ventur'd to ſay that the Magnus Orbis was but a Point in reſpect of the Sphere in which they

were placed, it was a plain Conſequence that every one of them that appeared any thing bright, muſt be larger than the Path or Orbit of the Earth: which is very abſurd. This is the principal Argument that Tycho Brahe ſet up againſt Copernicus. But when the Teleſcopes took away thoſe Rays of the Stars which appear when we look upon them with our naked Eye, (which they do beſt when the Eye-glaſs is black'd with Smoke) they ſeemed juſt like little ſhining Points, and then that Difficulty vaniſhed, and the Stars may yet be ſo many Suns. Which is the more probable, becauſe their Light is certainly their own: for it's impoſſible that ever the Sun ſhould ſend, or they reflect it at ſuch a vaſt Diſtance. This is the Opinion that commonly goes along with Copernicus's Syſtem. [They are not all in the ſame Sphere.] And the Patrons of it do alſo with reaſon ſuppoſe, that all theſe Stars are not in the ſame Sphere, as well becauſe there's no Argument for it, as that the Sun, which is one of them, cannot be brought to this Rule. But it's more likely they are ſcatter'd and diſpers'd all over the immenſe Spaces of the Heaven, and are as far diſtant perhaps from one another, as the neareſt of them are from the Sun.

Here again too I know Kepler is of another Opinion in his Epitome of Copernicus's Syſtem, that we mention'd above. For tho' he agrees with us, that the Stars are diffus'd through all the vaſt Expanſe of the Heavens, yet he cannot allow that they have as large an empty Space about them as our Sun has. For then 'twas his Opinion, we ſhould ſee but very few, and thoſe of very different Magnitudes: For, ſeeing the largeſt of all appear ſo ſmall to us, that we can ſcarce obſerve or meaſure them with our beſt Inſtruments; how muſt thoſe appear that are three or four times farther from us? Why, ſuppoſing them no larger than theſe, they muſt ſeem three or four times leſs, and ſo on 'till a little farther they will not be to be ſeen at all: Thus we ſhall have the ſight of but very few Stars, and thoſe very different one from another; Whereas we have above a Thouſand, and thoſe not conſiderably bigger or leſs than one another. But this by no means proves what he would have it; and his Miſtake was chiefly, that he did not conſider the Nature of Fire and Flame which may be ſeen at ſuch diſtances, and at ſuch ſmall Angles as all other Bodies would totally diſappear under. A thing that we need go no farther than the Lamps ſet along the Streets to prove. For altho' they are a hundred Foot from one another, yet you may count Twenty of them in a continued Row with your Eyes, and yet the twentieth Part of them ſcarce makes an Angle of ſix Seconds. Certainly then the glorious Light of the Stars muſt do much more than this; ſo that it's no wonder we ſhould ſee a Thouſand or two of them with our bare Eyes, and with a Teleſcope diſcover twenty times that number. But Kepler had a private Deſign in making the Sun thus ſuperiour to all the other Stars, and planting it in the Middle of the World, attended with the Planets: For his Aim was hereby to ſtrengthen his Coſmographical Myſtery, that the Diſtances of the Planets from the Sun are in a certain proportion to the Diameters of the Spheres that are inſcribed within, and circumſcribed about Euclid's Regular Bodies. Which could never be ſo much as probable, except there were but one Chorus of Planets moving round the Sun, and ſo the Sun were the only one of his kind.

But that whole Myſtery is nothing but an idle Dream taken from Pythagoras or Plato's Philoſophy. And the Author himſelf acknowledges that the Proportions do not agree ſo well as they ſhould, and is fain to invent two or three very ſilly Excuſes for it.

And he uſes yet poorer Arguments to prove that the Univerſe is of a ſpherical Figure, and that the Number of the Stars muſt neceſſarily be finite, becauſe the Magnitude of each of them is ſo. But what is worſt of all is, that he ſettles the Space between the Sun and the Concavity of the Sphere of the fix'd Stars, to be ſix hundred thouſand of the Earth's Diameters. For this reaſon, which he has no Foundation for, that as the Diameter of the Sun is to that of the Orbit of Saturn, which he makes to be as 1 to 2000, ſo is this Diameter to that of the Sphere of the fixed Stars. I cannot but wonder how ſuch things as theſe could fall from ſo ingenious a Man, and ſo great an Aſtronomer. But I muſt be of the ſame Opinion with all the greateſt Philoſophers of our Age, that the Sun is of the ſame Nature with the fix'd Stars. And this will give us a greater Idea of the World, than all thoſe other Opinions. [The Stars have Planets about them like our Sun.] For then why may not every one of theſe Stars or Suns have as great a Retinue as our Sun, of Planets, with their Moons, to wait upon them? Nay, there's a manifeſt reaſon why they ſhould. For if we imagine our ſelves placed at an equal diſtance from the Sun and fix'd Stars; we ſhould then perceive no difference between them. For, as for all the Planets that we now ſee attend the Sun, we ſhould not have the leaſt glimpſe of them, either becauſe their Light would be too weak to affect us, or that all the Orbs in which they move would make up one lucid Point with the Sun. In this Station we ſhould have no occaſion to imagine any difference between the Stars, and ſhould make no doubt if we had but the Sight, and knew the Nature of one of them, to make that the Standard of all the reſt. We are then plac'd near one of them, namely, our Sun, and ſo near as to diſcover ſix other Globes moving round him, ſome of them having others performing them the ſame Office. Why then may not we make uſe of the ſame Judgment that we would in that caſe; and conclude, that our Star has no better attendance than the others? So that what we allowed the Planets, upon the account of our enjoying it, we muſt likewiſe grant to all thoſe Planets that ſurround that prodigious number of Suns. They muſt have their Plants and Animals, nay and their rational Creatures too, and thoſe as great Admirers, and as diligent Obſervers of the Heavens as our ſelves; and muſt conſequently enjoy whatſoever is ſubſervient to, and requiſite for ſuch Knowledge.

What a wonderful and amazing Scheme have we here of the magnificent Vaſtneſs of the Univerſe! So many Suns, ſo many Earths, and every one of them ſtock'd with ſo many Herbs, Trees, and Animals, and adorn'd with ſo many Seas and Mountains! And how muſt our Wonder and Admiration be increaſed when we conſider the prodigious Diſtance and Multitude of the Stars?

That their Diſtance is ſo immenſe, that the Space between the Earth and Sun (which is no leſs than Twelve thouſand of the Earth's Diameters) is almoſt nothing when compar'd to it, has more Proofs than one to confirm it. And this among the reſt. If you obſerve two Stars near one another, as for example thoſe in the middle of the Great Bears Tail, differing very much from one another in Clearneſs, notwithſtanding our changing our Poſition in our Annual Orbit round the Sun, and that there would be a Parallax were the Star which is brighter nearer to us than the other, as is very probable it is, yet whatever Part of the Year you look upon them, they will not in the leaſt have altered their diſtance. Thoſe that have hitherto undertook to calculate their Diſtance, have not been able perfectly to compaſs their Deſign, by reaſon of the

extreme Niceneſs and almoſt Impoſſibility of the Obſervations requiſite for their Purpoſe. The only Method that I ſee remaining, to come at any tolerable Probability in ſo difficult a Caſe, I ſhall here make uſe of. Seeing then that the Stars, as I ſaid before, are ſo many Suns, if we do but ſuppoſe one of them equal to ours, it will follow that its diſtance from us is as much greater than that of the Sun, as its apparent Diameter is leſs than the Diameter of the Sun. But the Stars, even thoſe of the firſt Magnitude, though view'd through a Teleſcope, are ſo very ſmall, that they ſeem only like ſo many ſhining Points, without any perceivable Breadth. So that ſuch Obſervations can here do us no good. When [A way of making a probable gueſs at the diſtance of the Stars.] I ſaw this would not ſucceed, I ſtudied by what way I could ſo leſſen the Diameter of the Sun, as to make it not appear larger than the Dog, or any other of the chief Stars. To this purpoſe I clos'd one End of my twelve-foot Tube with a very thin Plate, in the Middle of which I made a Hole not exceeding the twelfth Part of a Line, that is the hundred and forty fourth Part of an Inch. That End I turn'd to the Sun, placing my Eye at the other, and I could ſee ſo much of the Sun as was in Diameter about the 182d part of the Whole. But ſtill that little piece of him was brighter much than the Dog-ſtar is in the cleareſt Night. I ſaw that this would not do, but that muſt leſſen the Diameter of the Sun a great deal more. I made then ſuch another Hole in a Plate, and againſt it I plac'd a little round Glaſs that I had made uſe of in my Microſcopes, of much about the ſame Diameter with the former Hole. Then looking again towards the Sun (taking care that no Light might come near my Eye to hinder my Obſervation) I found it appeared of much the ſame Clearneſs with Sirius. But caſting up my account, according to the Rules of Dioptricks, I found his Diameter now was but 1⁄152 part of that hundred and eighty ſecond part of his whole Diameter that I ſaw through the former Hole. Multiplying 1⁄152 and 1⁄182 into one another, the Product I found to be 1⁄27664. The Sun therefore being contracted into ſuch a Compaſs, or being removed ſo far from us (for it's the ſame thing) as to make his Diameter but the 27664 part of that we every Day ſee, will ſend us juſt the ſame Light as the Dog-ſtar now doth. And his diſtance then from us will be to his preſent diſtance undoubtedly as 27664 is to 1; and his Diameter little above four Thirds, 4‴. Seeing then Sirius is ſuppoſed equal to the Sun, it follows that his Diameter is likewiſe 4‴, and that his Diſtance to the Diſtance of the Sun from us is as 27664 to 1. And what an incredible Diſtance that is, will appear by the ſame way of reaſoning that we uſed in meaſuring that of the Sun. For if 25 Years are required for a Bullet out of a Cannon, with its utmoſt Swiftneſs, to travel from the Sun to us; then by multiplying the Number 27664 into 25, we ſhall find that ſuch a Bullet would ſpend almoſt ſeven hundred thouſand Years in its Journey between us and the neareſt of the fix'd Stars. And yet when in a clear Night we look upon them, we cannot think them above ſome few Miles over our Heads. What I have here enquir'd into, is concerning the neareſt of them. And what a prodigious Number muſt there be beſides of thoſe which are placed in the vaſt Spaces of Heaven, as to be as remote from theſe as theſe are from the Sun! For if with our bare Eyes we can obſerve above a Thouſand, and with a Teleſcope can diſcover ten or twenty times as many; what bounds of Number can we ſet to thoſe which are out of the Reach even of theſe Affiſtances! eſpecially if we conſider the infinite Power of God. Really, when I have been reflecting thus with my ſelf, me-thoughts all our Arithmetick was nothing, and we are vers'd but in the very Rudiments of Numbers, in

compariſon of this great Sum. For this requires an immenſe Treaſury, not of twenty or thirty Figures only, in our decuple Progreſſion, but of as many as there are Grains of Sand upon the Shore. And yet who can ſay, that even this Number exceeds that of the Fix'd Stars? Some of the Ancients, and Jordanus Brunus carry'd it further, in declaring the Number infinite: he would perſwade us that he has prov'd it by many Arguments, tho' in my opinion they are none of them concluſive. Not that I think the contrary can ever be made out. Indeed it ſeems to me certain, that the Univerſe is infinitely extended; but what God has been pleas'd to place beyond the Region of the Stars, is as much above our Knowledge, as it is beyond our Habitation.

Or what if beyond ſuch a determinate Space he has left an infinite Vacuum; to ſhow, how inconſiderable all that he has made is, to what his Power could, had he ſo pleas'd, have produced? But I am falling, before I am aware, into that intricate Diſpute of Infinity: Therefore I ſhall wave this, and not, as ſoon as I am free of one, take upon me another difficult Task. All that I ſhall do more is to add ſomewhat of my Opinion concerning the whole World, as it is a Place for the Reception of the Suns or fix'd Stars, every one of which, I have ſhowed, may have their Planetary Syſtems about them.

[Every Sun has a Vortex round it, very different from thoſe of Cartes.]

I am of Opinion then that every Sun is ſurrounded with a Whirl-pool or Vortex of Matter in a very ſwift Motion; tho' not in the leaſt like Cartes's either in their Bulk, or manner of Motion. For Cartes makes his ſo large, as every one of them to touch all the others round them, in a flat Surface, juſt as you have ſeen the Bladders that Boys blow up in Soap-ſuds do; and would have the whole Vortex to move round the ſame way. But the Angles of every Vortex will be no ſmall hindrance to ſuch a Motion. Then the whole Matter moving round at once, upon the Axis as it were of a Cylinder, did not a little puzzle him in giving Reaſons for the Roundneſs of the Sun: which however they may ſatisfy ſome People that do not conſider them, really prove nothing of the Matter. In this æthereal Matter the Planets float, and are carried round by its Motion: and the thing that keeps them in their own Orbs is, that they themſelves, and the Matter in which they ſwim, equally ſtrive to fly off from the Center of this Motion. Againſt all which there are many Aſtronomical Objections, ſome of which I touch'd upon in my Eſſay of the Cauſes of Gravity. Where I gave another Account of the Planets not deſerting their own Orbs; which is their Gravitation towards the Sun. I ſhow'd there the Cauſes of that Gravitation, and cannot but wonder that Cartes, the firſt Man that ever began to talk reaſonably of that Matter, ſhould never meddle with, or light on it. Plutarch in his Book of the Moon above-mentioned ſays, that ſome of the Ancients were of Opinion, that the Reaſon of the Moon's keeping her Orbit was, that the Force of her Circular Motion was exactly equal to her Gravity, the one of which pull'd her to, as much as the other forc'd off from the Centre. And in our Age Alphonſus Borellus, who was of this ſame Opinion in the other Planets as well as the Moon, makes the Gravitation of the primary Planets to be towards the Sun, as that of the Secondary is towards the Planets round which they move: Which Sir Iſaac Newton has more fully explain'd, with a great deal of Pains and Subtilty; and how from that Cauſe proceeds the Ellipticity of the Orbs of the Planets, found out by Kepler.

According to my Notion of the Gravitation of the Planets to the Sun, the Matter of his Vortex muſt not at all move the ſame way, but after ſuch a manner as to have its Parts carry'd different ways on all Sides. And yet there is no fear of its being deſtroyed by ſuch an irregular Motion, becauſe the Æther round it, which is at reſt, keeps the Parts of it from flying out. With the Help of ſuch a Vortex as this I have undertook in that Eſſay to explain the Gravity of Bodies on this Earth, and all the Effects of it. And I ſuppoſe there may be the ſame Cauſe as well of the Gravitation of the Planets, and of our Earth among the reſt, towards the Sun, as of their Roundneſs: A Thing ſo very hard to give an Account of in Cartes's Syſtem.

I muſt differ from him too in the Bigneſs of the Vortices, for I cannot allow them to be ſo large as he would make them. I would have them diſperſed all about the immenſe Space, like ſo many little Whirl-pools of Water, that one makes by the ſtirring of a Stick in any large Pond or River, a great way diſtant from one another. And as their Motions do not all intermix or communicate with one another, ſo in my Opinion muſt the Vortices of Stars be placed as not to hinder one anothers free Circumrotations.

So that we may be ſecure, and never fear that they will ſwallow up or deſtroy one another; for that was a mere Fancy of Cartes's, when he was a ſhowing how a fix'd Star or Sun might be turn'd into a Planet. And 'tis plain that when he writ it, he had no Thoughts of the immenſe Diſtance of the Stars from one another; particularly, by this one Thing, that he would have a Comet as ſoon as ever it comes into our Vortex, to be ſeen by us. Which is as abſurd as can be. For how could a Star, which gives us ſuch a vaſt Light only from the Reflection of the Beams of the Sun, as he himſelf owns they do; how I ſay could that be ſo plainly ſeen at a diſtance Ten thouſand times larger than the Diameter of the Earth's Orbit? He could not but know that all round the Sun there is a vaſt Extenſum; ſo vaſt, that in Copernicus's Syſtem the magnus Orbis is counted but a Point in compariſon with it. But indeed all the whole Story of Comets and Planets, and the Production of the World, is founded upon ſuch poor and trifling Grounds, that I have often wonder'd how an ingenious Man could ſpend all that pains in making ſuch Fancies hang together. For my part, I ſhall be very well contented, and ſhall count I have done a great Matter, if I can but come to any Knowledge of the Nature of Things, as they now are, never troubling my ſelf about their Beginning, or how they were made, knowing that to be out of the reach of human Knowledge, or even Conjecture.

FINIS.

NOTE

[1] The Author invented the Pendulum for Clocks.